Jahangirnagar University
Department of Chemistry

Lab Report

Course Title: Physical Chemistry Practical (III)

Course No: CHEM - 315

Experiment No: 01

Experiment Name: Determination of the Molecular Weight of a Substance by Cryoscopic Method.

Submitted To

Dr. Mohammed Abdullah Khan
Professor
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Submitted By:

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Session: 2020-2021
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Experiment NO: 1

Name of experiment: Determination the molecular weight of a substance by cryoscopic method.

Theory: The properties of solution differ from those of a pure solvent due to interactions that take place btween the solute and solvent molecules. These properties are depend only upon the number of particles that are dissolved in the solvent and not on the identity of the particles. when a particular solute is dissolved, the following expression holds true:

$$\Delta T = T_f^\circ - T_f = K_f m = K_f \frac{1000\, W_1}{M W_2} \quad \text{---} \ (1)$$

The term T_f° and T_f refer to the freezing point temperatures of the pure solvent and

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the solution, respectively. The constant, K_f is referred to as the freezing-point-depression constant and is depend only upon the solvent. For cyclo hexane, $K_f = 20.2$ deg. kg/mol of solute.

Apparatus and chemical required:

1. Digital thermometer (−10° to 100°C)
2. Stirring rod
3. ice
4. cyclohexane
5. camphor
6. Stopwatch etc.

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Experiment NO.01

Name of the experiment: Determination of the molecular weight of a substance by cryoscopic method.

DATA: Table for the determination of freezing point of pure solvent (cyclohexane).

Time(min)	Temp(°C)	Time(min)	Temp(°C)	Time(min)	Temp(°C)
0	32.0	7.0	19.2	14.0	10.6
0.5	31.6	7.5	18.9	14.5	9.9
1.0	30.8	8.0	18.4	15.0	9.6
1.5	29.9	8.5	17.9	15.5	9.1
2.0	29.1	9.0	17.5	16.0	8.4
2.5	28.1	9.5	16.8	16.5	8.0
3.0	27.0	10.0	16.0	17.0	7.5
3.5	25.8	10.5	15.2	17.5	7.2
4.0	25.2	11.0	14.5	18.0	7.0
4.5	24.1	11.5	13.1	18.5	6.6
5.0	23.5	12.0	12.5	19.0	6.6
5.5	22.9	12.5	11.9	19.5	6.6
6.0	21.4	13.0	11.5	20.0	6.6
6.5	20.6	13.5	11.0	20.5	6.6

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Table for determination of freezing point of the mixture of cyclohexane and camphor.

Time (min)	Temperature (°C)	Time (min)	Temperature (°C)
0.0	30.5	12.5	11.0
0.5	29.9	13.0	10.7
1.0	29.1	13.5	10.1
1.5	28.2	14.0	9.5
2.0	27.7	14.5	8.9
2.5	27.1	15.0	8.5
3.0	26.0	15.5	8.0
3.5	25.0	16.0	7.9
4.0	23.6	16.5	6.9
4.5	22.9	17.0	6.5
5.0	22.3	17.5	5.8
5.5	21.2	18.0	5.5
6.0	20.1	18.5	5.1
6.5	19.3	19.0	5.0
7.0	18.7	19.5	4.8
7.5	17.9	20.0	4.6
8.0	17.1	20.5	4.1
8.5	16.3	21.0	3.9
9.0	15.9	21.5	3.7
9.5	14.9	22.0	3.4
10.0	14.9	22.5	3.1
10.5	13.8	23.0	3.1
11.0	13.1	23.5	3.1
11.5	12.9		
12.0	11.6		

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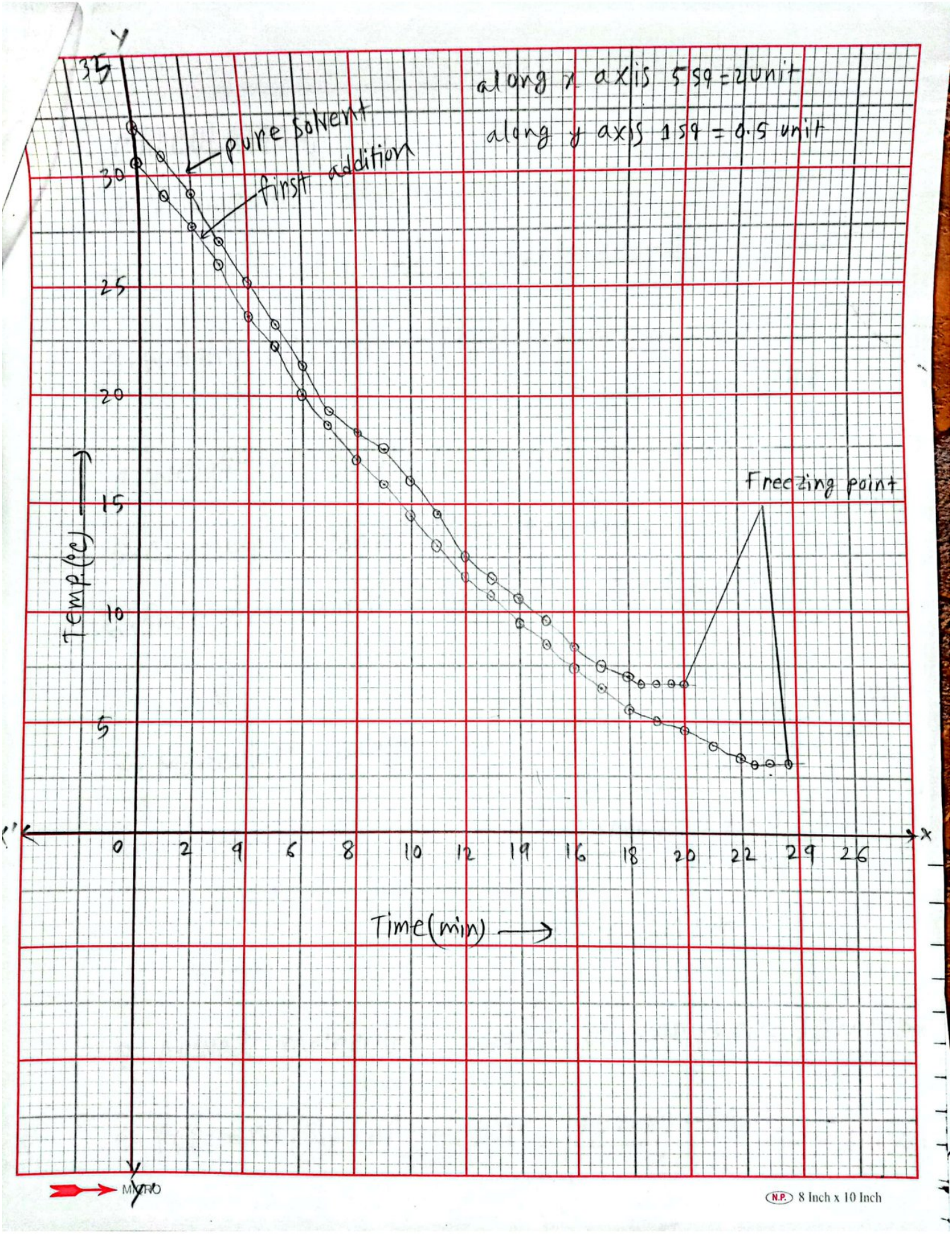

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CALCULATION:

1. Volume of solvent (cyclohexane) = 25.00ml

2. Density of solvent (cyclohexane) = $0.79\ g\ ml^{-1}$

3. weight of solvent (cyclohexane), $W_2 = (25 \times 0.79)g$
$= 19.75g$

4. weight of solute (camphor), $W_1 = 0.50g$

5. cryoscopic constant of solvent, $K_f = 20.2\ deg.kg\ mol^{-1}$

6. freezing point depression, $\Delta T = (6.6 - 3.1)^\circ C$
$= 3.5^\circ C$

7. Molecular weight of camphor, $M_c = \frac{1000 \times K_f \times W_1}{W_2 \times \Delta T}$

$= \frac{1000 \times 20.2 \times 0.50}{19.75 \times 3.5}$

$= 146.11g$

8. Actual molecular weight of camphor = $152.23\ g\ mol^{-1}$

9. percent (%) of error = $\left|\frac{152.23 - 146.11}{152.23}\right| \times 100\%$

$= 4.02\%$

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RESULT:

1. Molecular weight of substance (camphor) is 146.11 g.

2. percent (%) of error is 4.02%.

DISCUSSION:

1. The freezing point of the solution is less than that of pure solvent. This is thermodynamic factor.
2. In this experiment, cyclohexane used as solvent and camphor used as solute.
3. Measurement was taken every 30 second carefully.
4. Air Jacket was used outside the glass tube containing solvent or solution for shaking of uniform cooling.
5. Freezing point deppression occurs due to colligative properties.

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Jahangirnagar University
Department of Chemistry

Lab Report

Course Title: Physical Chemistry Practical (III)

Course No: CHEM - 315

Experiment No: 02

Experiment Name: Study of the adsorption of acetic acid on charcoal and examination of the validity of Freundich's adsorption isotherm and Langmuirs adsorption isotherm.

Submitted By:

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Roll: 200

Group No: 09

Session: 2020-2021

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Experiment NO: 02

Name of the experiment: studing the adsorption of acetic acid on charcoal and examination the validity of freundlich's adsorption isotherm and langmuirs adsorption isotherm.

Theory: The amound of adsorbed by any adsorbent is dependent on temperature and pressure. The amound of adsorbed is a function of pressure and temperature, i.e.

$$X = f(P,T)$$

Freundlich adsorption isotherm: Freundlich adsorption isotherm is represented by,

$$\frac{x}{m} = KC^{1/n}$$

where x is the amount of solute adsorbed, m is the amount of adsorbing material, c is the equilibrium concentration of adsorbate in

the solution, K is a constant depending upon the nature of both adsorbant and adsorbate. on taking logarithm of eqn ①,

$$\log \frac{x}{m} = \log K + \frac{1}{n} \log C$$

langmuir adsorption isotherm: the following relation holds for adsorption

$$\frac{x}{m} = \frac{K_1 K_2 C}{1 + K_1 C}$$

where K_1 and K_2 are constant. The above eqn can be written as

$$\frac{C}{x/m} = \frac{1}{K_1 K_2} + \frac{C}{K_2}$$

Thus a plot of $\frac{C}{x/m}$ against C gives a straight line with a slope of $1/K_1$ and intercept of $1/K_1K_2$ and shows the validity of langmuir's adsorption isotherm.

Apparatus and chemical required:

Burette, pipette, activated charcoal, 0.5N acetic acid, 0.1N sodium hydroxide, stoppered bottles, shaking machine, phenophthelien indicator.

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Experiment NO: 02

Name of Experiment: studing the adsorption of acetic acid on charcoal & examining the validity of freundlich's adsorption isotherm and langmuirs adsorption isotherm.

DATA: Experimental data table for titration of acetic acid solution with 0.1N NaOH.

Bottle-1:

No. of obs.	Volume of CH_3COOH (ml)	Volume of 0.1N NaOH Solution			
		I.B.R.	F.B. reading	Difference	Mean
1	10.00	0.00	49.6	49.6	49.53
2	10.00	0.00	49.5	49.5	
3	10.00	0.00	49.5	49.5	

Bottle-2:

No. of obs.	Volume of CH_3COOH (ml)	Volume of 0.1N NaOH Solution			
		I.B.R.	F.B.R.	Difference	Mean
1	10.00	0.00	38.5	38.5	
2	10.00	0.00	38.6	38.6	38.5
3	10.00	0.00	38.5	38.5	

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Bottle-3

No of obs.	Volume of CH_3COOH (ml)	Volume of 0.1N NaOH Solution			
		I.B.R	F.B.R.	Difference	Mean
1	10.0	0.00	27.8	27.8	27.9
2	10.0	0.00	27.9	27.9	
3	10.0	0.00	27.9	27.9	

Bottle-4

No. of obs.	Volume of CH_3COOH (ml)	Volume of 0.1N NaOH solution			
		I.B.R	F.B.R.	Difference	Mean
1	10.0	0.00	22.7	22.7	22.8
2	10.0	0.00	22.8	22.8	
3	10.0	0.00	22.8	22.8	

Bottle-5

No. of obs.	Volume of CH_3COOH (ml)	Volume of 0.1N NaOH Solution			
		I.B.R.	F.B.R.	Difference	Mean
1	10.0	0.00	17.7	17.7	17.7
2	10.0	0.00	17.8	17.8	
3	10.0	0.00	17.7	17.7	

Bottle-6

No. of obs.	Volume of CH_3COOH (ml)	Volume of 0.1N NaOH Solution			
		I.B.R.	F.B.R.	Difference	Mean
1	10.0	0.00	8.3	8.3	8.3
2	10.0	0.00	8.4	8.4	
3	10.0	0.00	8.3	8.3	

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CALCULATION:

(a) Determination of amount of acetic acid adsorbed:

Bottle No	Initial concentration of acid (C_o) before adsorbed	Equilibrium concentration (C_e) of acid after adsorption	amount of acid ($C_o - C_e$) adsorbed (ml of NaOH) = X
1	50	49.5	0.47
2	40	38.5	1.47
3	30	27.9	2.13
4	25	22.8	2.23
5	20	17.7	2.27
6	10	8.3	1.67

(b) Data for validity of Freundlich's and langmuir adsorption isotherm:

Bottle No.	C_e	x/m	$\log C_e$	$\log x/m$	$\frac{C_e}{x/m}$
1	49.5	0.47	1.69	−0.33	105.54
2	38.5	1.47	1.59	0.17	26.21
3	27.9	2.13	1.45	0.33	13.08
4	22.8	2.23	1.36	0.35	10.24
5	17.7	2.27	1.25	0.36	7.87
6	8.3	1.67	0.92	0.22	4.99

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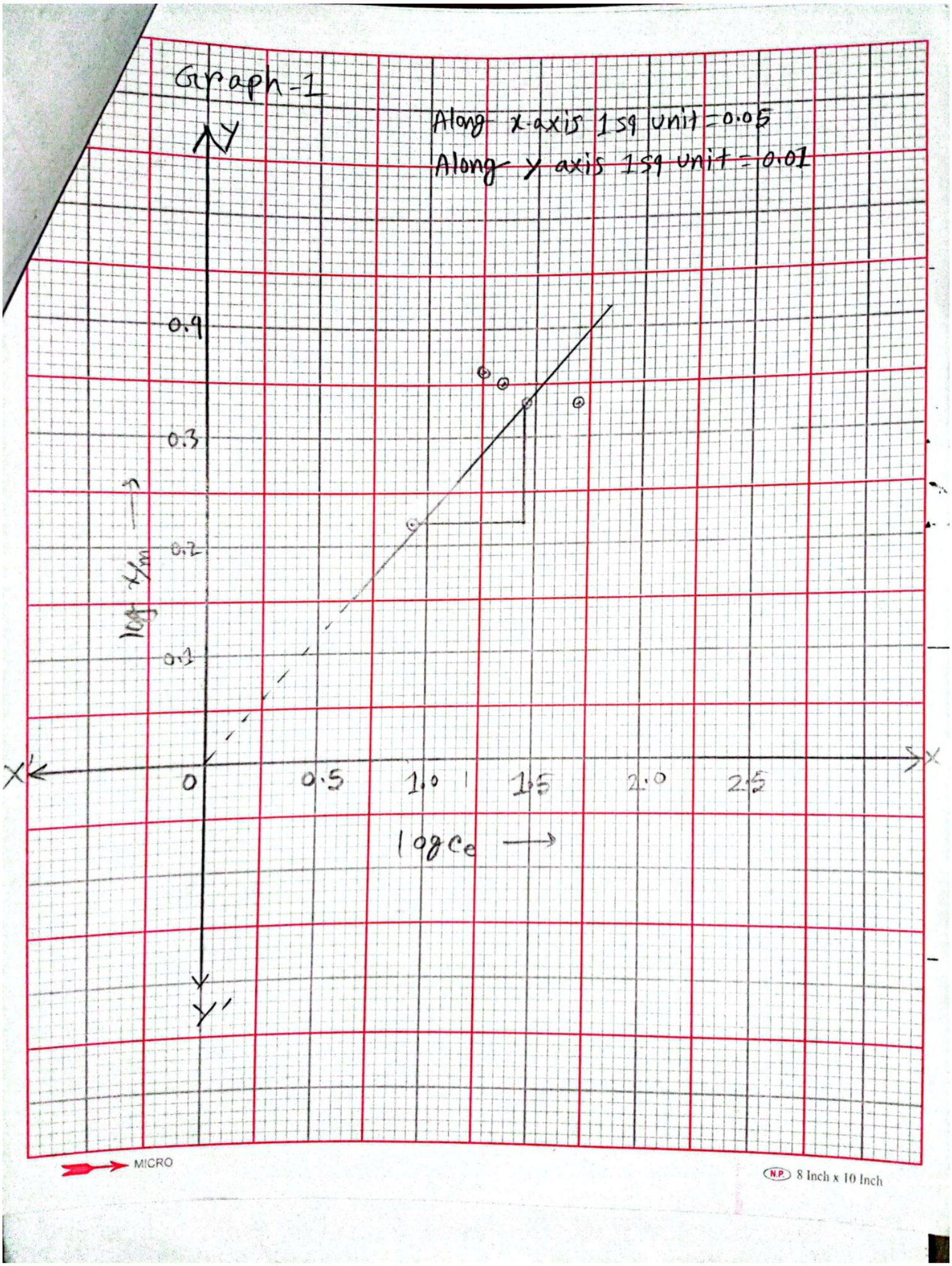

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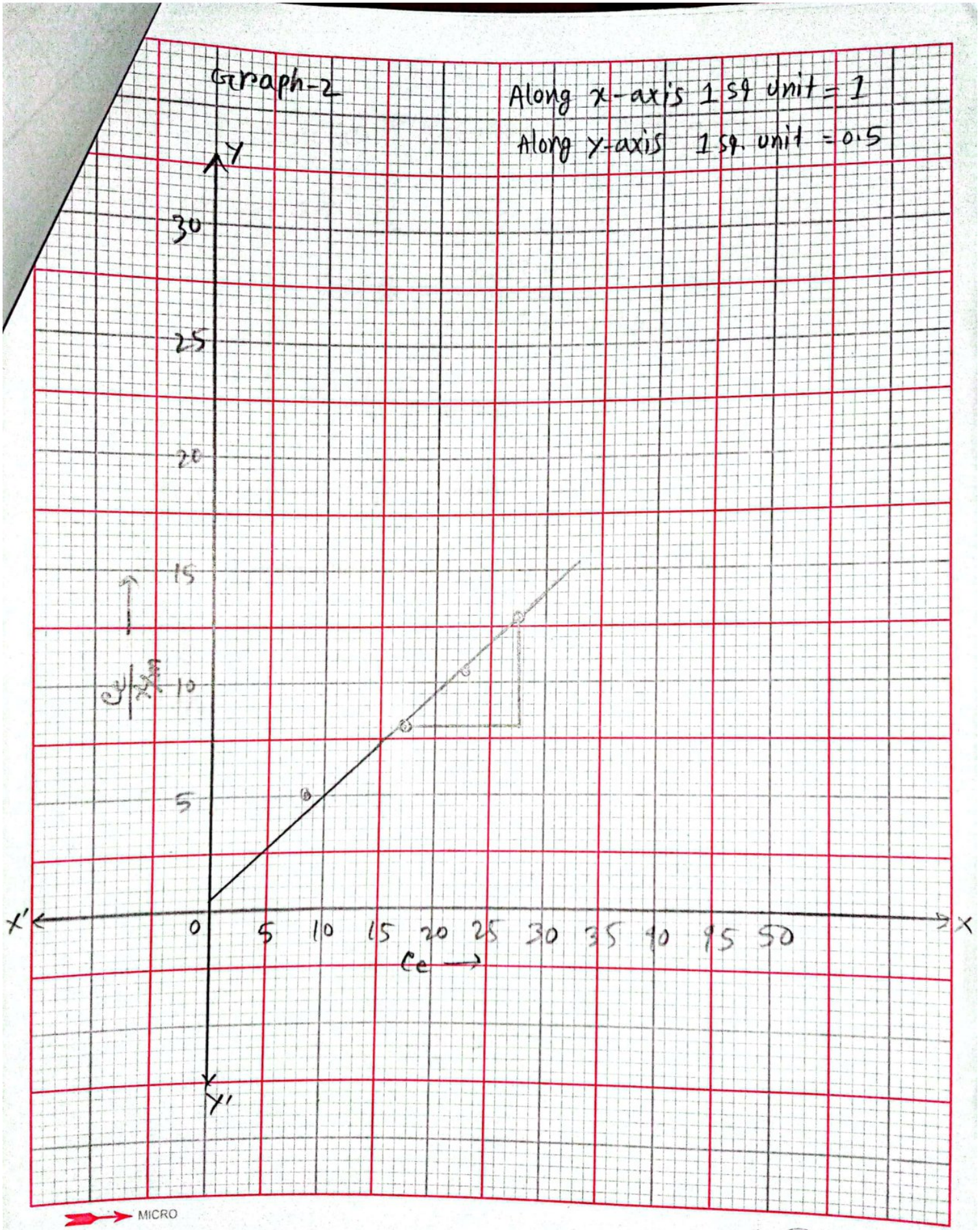
Graph-2

Along x-axis 1 sq unit = 1

Along y-axis 1 sq. unit = 0.5



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For Freundlich adsorption isotherm:

From graph-1,

$$\text{slope} = \frac{0.33 - 0.22}{1.95 - 0.92}$$

$$\Rightarrow \frac{1}{n} = 0.21$$

the intercept $\log K = 0$

For Langmuir adsorption isotherm:

$$\frac{C}{x/m} = \frac{1}{K_1 K_L} + \frac{C}{K_L}$$

From graph-2

$$\text{Slope} = \frac{13.08 - 7.81}{27.87 - 17.73}$$

$$\therefore \frac{1}{K_2} = 0.52$$

$$\therefore K_L = 1.92$$

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RESULT: The validity of Freundlich adsorption isotherm is studied by plotting of log x/m vs log C_e in graph-1 which gives a straight line with slope $\frac{1}{n} = 0.21$, intercept $\log K = 0$.

The validity of Langmuir method is studied up plotting $\frac{C_e}{x/m}$ vs C_e in graph-2 which give a straight line and $K_L = 1.92$ and intercept 0.50.

Discussion:

1. For titration small Filter paper was used to minimize the error due to the adsorption of acid by filter paper.
2. The first 5 ml Filtrate of each bottle was thrown away For accurate solution.
3. The weight & readings were taken carefully.
4. At begining of exp. & after being adsorbed, I determined concentration of acetic acid.
5. By using acetic acid and indicator I determine conc. of NaOH

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Jahangirnagar University
Department of Chemistry

Lab Report

Course Title: Physical Chemistry Practical (III)

Course No: CHEM - 315

Experiment No: 03

Experiment Name: Study of the liquid-liquid miscibility of three liquid system and determination of the position of one tie line.

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Submitted By:

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Group No: 09
Session: 2020-2021
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Experiment NO: 3

Name of experiment: studing the liquid-liquid miscibility of three liquid system and determination of the position of one tie line.

Theory: Liquid system of three components consist of three liquids components, two liquids and one solid or one liquid and two solids. In this experiment we investigate the behaviour of a system of three liquids : $CHCl_3$, CH_3COOH and H_2O. Acetic acid (CH_3COOH) is miscible with both dichloromethane and water but $CHCl_3$ and H_2O are quit insoluble in each other.

According to the phase rule, a single phase in a three component system may posses four degree of freedom.

$$F = C - P + 2 = 3 - 1 + 2 = 4$$

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where, F = degree of freedom, c = component

P = phase

These are temperature, pressure and the compositions of two of the three components. Because of the difficulty in graphically so many variables, temperature and pressure are generally held constant, so the phase rule reduces to, $F = c - P = 3 - P$

This is the same special form of the phase rule that applies to two component systems of constant pressure only. graphical represent of a three component system:

For a three component system, the properties of an equilateral triangle provide a convenient means of representing the composition of three component system at constant temperature and pressure. line AB represent the

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compositions of all systems that can be prepared from pure A and pure B, at point P the system is 60% B and 40% A.

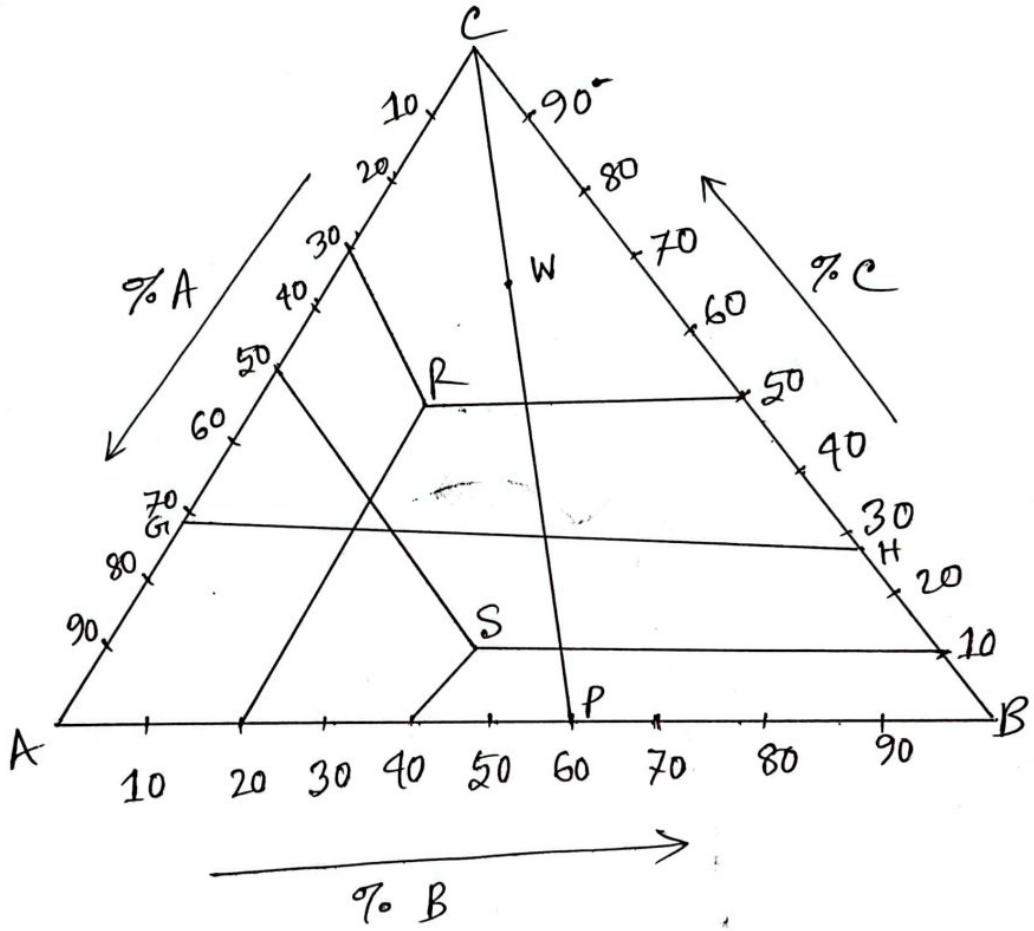

Fig. 1- Graphical representation of a three component system at constant pressure and temperature.

In Fig. 1, the edges of the triangle represent the compositions of all the two component systems that can be prepared from A and B, B and C,

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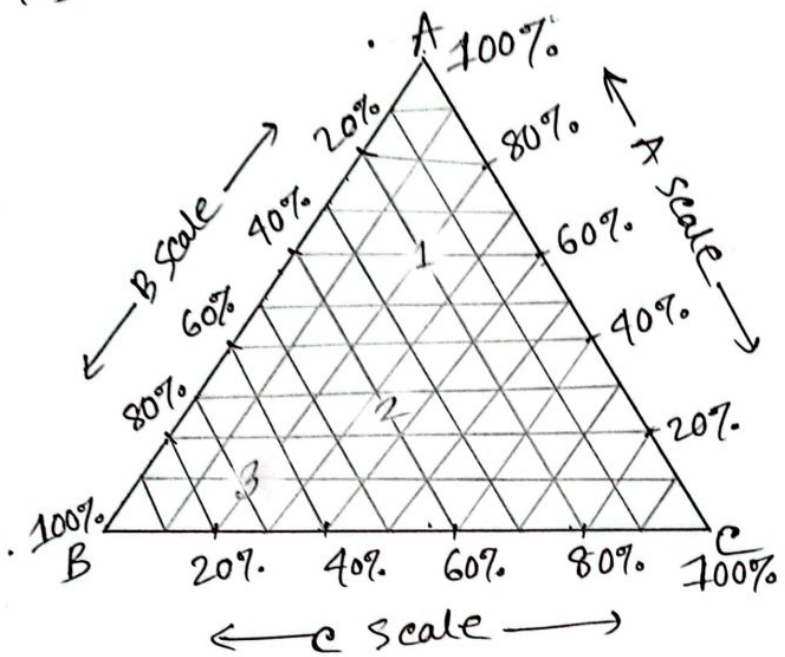
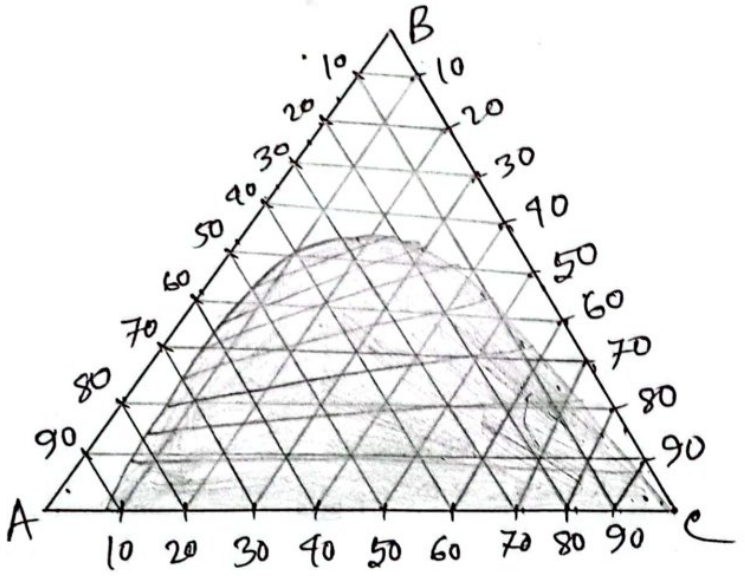
and A and C. At point R the system is 50% C, 20% B and 30% A. At point S the system is 10% C, 40% B and 50% A.





Apparatus and chemical required:

1. Five stoppered bottles
2. Burette . 3. Chloroform
4. water etc .

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Experiment NO:03

NAME OF EXPERIMENT: Studing the liquid-liquid miscibility of three liquid system and determination of the position of one tie line.

DATA:

% of $CHCl_3$	weight of $CHCl_3$ (gm)	volume of $CHCl_3$, $V=\frac{W}{\rho}$ (ml)	weight of CH_3COOH (gm)	volume of CH_3COOH, $V=\frac{W}{\rho}$ (ml)	volume of H_2O (ml)
10%	2	1.33	18	17.14	27.87
20%	4	2.67	16	15.24	13.00
30%	6	4.00	14	13.33	7.20
40%	8	5.33	12	11.43	5.20
50%	10	6.67	10	9.52	3.10
60%	12	8.00	8	7.62	1.90
70%	14	9.33	6	5.71	0.90
80%	16	10.67	4	3.81	0.50
90%	18	12.00	2	1.90	0.30

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CALCULATION:

Determination of the percentage of $CHCl_3$, CH_3COOH and H_2O in the mixture.

mixture / Bottle	Total volume	% of $CHCl_3$	% of CH_3COOH	% of H_2O
10% $CHCl_3$ + 90% CH_3COOH	1.33+17.19+27.87 = 46.39	$\frac{1.33 \times 100}{46.39}$ = 2.87	$\frac{17.19 \times 100}{46.39}$ = 36.99	$\frac{27.87 \times 100}{46.39}$ = 60.14
20% $CHCl_3$ + 80% CH_3COOH	2.67+15.27+13.0 = 30.99	$\frac{2.67 \times 100}{30.99}$ = 8.63	$\frac{15.27 \times 100}{30.99}$ = 49.35	$\frac{13 \times 100}{30.99}$ = 42
30% $CHCl_3$ + 70% CH_3COOH	4.0+13.33+7.2 = 24.52	$\frac{4 \times 100}{24.52}$ = 16.3	$\frac{13.33 \times 100}{24.52}$ = 54.36	$\frac{7.2 \times 100}{24.52}$ = 29.36
40% $CHCl_3$ + 60% CH_3COOH	5.33+11.43+5.2 = 21.96	$\frac{5.33 \times 100}{21.96}$ = 24.27	$\frac{11.43 \times 100}{21.96}$ = 52.09	$\frac{5.2 \times 100}{21.96}$ = 23.67
50% $CHCl_3$ + 50% CH_3COOH	6.67+9.52+3.1 = 19.29	$\frac{6.67 \times 100}{19.29}$ = 34.73	$\frac{9.52 \times 100}{19.29}$ = 49.85	$\frac{3.1 \times 100}{19.29}$ = 16.07
60% $CHCl_3$ + 40% CH_3COOH	8+7.62+1.9 = 17.52	$\frac{8 \times 100}{17.52}$ = 45.66	$\frac{7.62 \times 100}{17.52}$ = 43.5	$\frac{1.9 \times 100}{17.52}$ = 10.89
70% $CHCl_3$ + 30% CH_3COOH	9.33+5.71+0.9 = 15.99	$\frac{9.33 \times 100}{15.99}$ = 58.53	$\frac{5.71 \times 100}{15.99}$ = 35.82	$\frac{0.9 \times 100}{15.99}$ = 5.69
80% $CHCl_3$ + 20% CH_3COOH	10.67+3.81+0.5 = 14.98	$\frac{10.67 \times 100}{14.98}$ = 71.23	$\frac{3.81 \times 100}{14.98}$ = 25.43	$\frac{0.5 \times 100}{14.98}$ = 3.33
90% $CHCl_3$ + 10% CH_3COOH	12+1.90+0.3 = 14.20	$\frac{12 \times 100}{14.2}$ = 84.51	$\frac{1.90 \times 100}{14.2}$ = 13.38	$\frac{0.3 \times 100}{14.2}$ = 2.11

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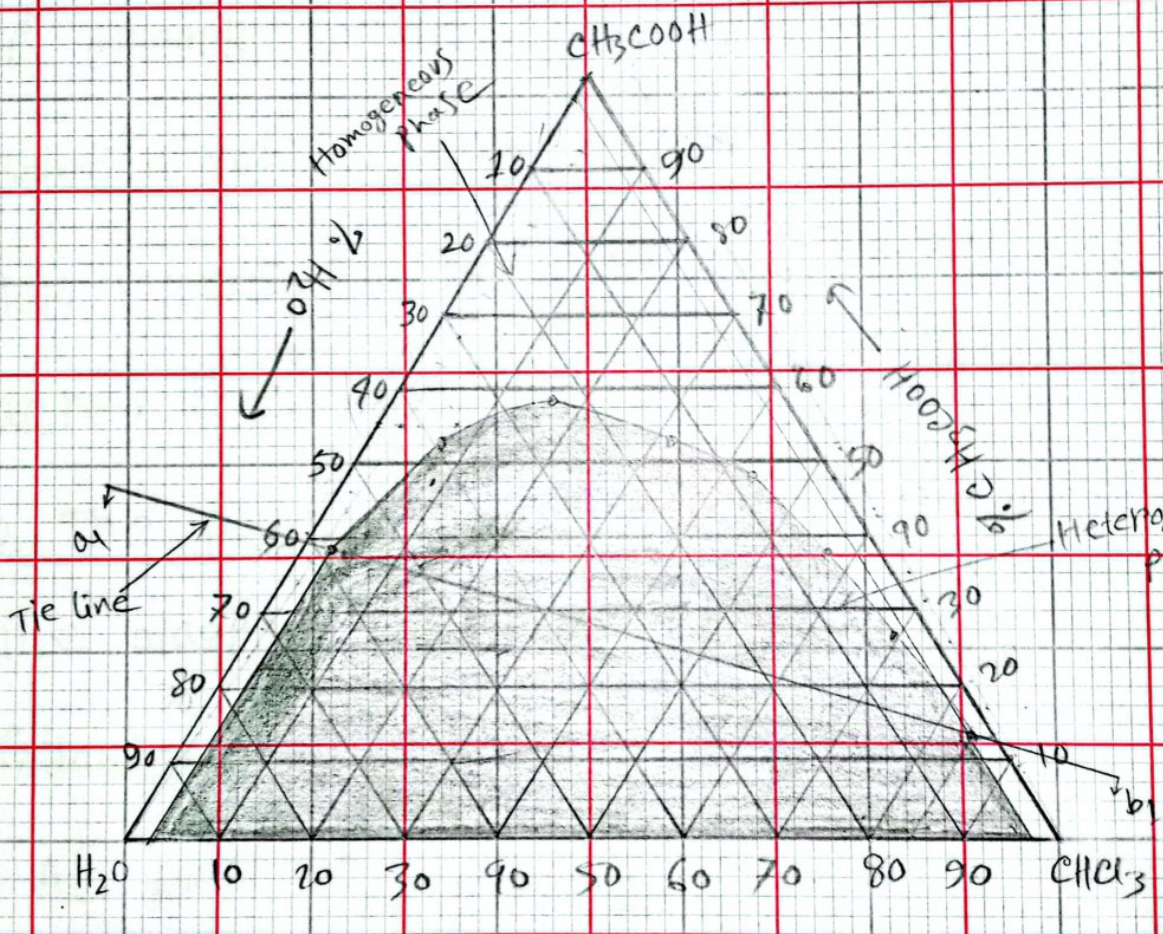

Fig : The tertiary phase diagram of three component ($CHCl_3$, CH_3COOH, H_2O) system.

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RESULT:

1. curve obtain from joining experimentally determined point from the boundary between homogeneous and heterogeneous mixture. A mixture of CH_3COOH inside the triangle will form homogenea solution while any mixture represented in two layers. It is required to draw the line across the miscibility indicate solution pair

2. The position of tie line is represented in the diagram. The point corresponding to the following composition,

At point a_1, % of $CHCl_3$ = 2.87
% of CH_3COOH = 36.99
% of H_2O = 60.14

At point b_1, % of $CHCl_3$ = 84.51
% of CH_3COOH = 13.38
% of H_2O = 2.11

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DISCUSSION:

1. Before the experiment, all the apparatus were cleaned carefully.

2. $CHCl_3$ is a volatile compound, so, during preparation of the mixture of $CHCl_3$ and CH_3COOH, some amount of $CHCl_3$ may vaporised. so we may not find the exact volume of $CHCl_3$.

3. In this experiment we have used glacial acetic acid, because glacial acetic acid do not contain any water. There all apparatus was moisture free.

4. Before starting the experiment, we dried all the instrument properly.

5. In titration the turbidity of solution was observed similarly change.

6. Temperature and pressure were kept constant.

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Jahangirnagar University
Department of Chemistry

Lab Report

Course Title: Physical Chemistry Practical (III)

Course No: CHEM - 315

Experiment No: 04

Experiment Name: Determination of the partial molal volume of solutions (Alcohol-Water system) by Westphal balance.

Submitted By:

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Session: 2020-2021

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Experiment NO: 04

Name of experiment: Determination the partial molal volume of solutions (alcohol-water system) by westphal balance.

THEORY: The partial molal property of a component of a solution may be defined as follows: Let G represent any extensive property of a binary solution; at constant temperature and pressur, G then will be a function of the two independent variables n_1 and n_2 which represent the number of moles of component 1 and component 2 respectively. The partial molal quantity of component 1 is then defined by the relation-

$$\bar{G}_1 = \left(\frac{\delta G}{\delta n_1}\right)_{n_2, T, P} \quad \text{-----} (1)$$

similarly for component 2,

$$\bar{G}_2 = \left(\frac{\delta G}{\delta n_2}\right)_{n_1, T, P} \quad \text{---} (2)$$

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Since any extensive property G of a mixture is a homogeneous function of ~~degree~~ one of the two independent variables (n_1 & n_2),

$$G(n_1 \text{ and } n_2) = n_1\bar{G}_1 + n_2\bar{G}_2 \quad (T, P \text{ constant}) \text{ -- } ③$$

In the case of the volume of the solution by similar reasoning we have –

$$V(n_1 \text{ and } n_2) = n_1\bar{V}_1 + n_2\bar{V}_2 \quad (T, P \text{ constant}) \text{ ---- } ④$$

For determining $\bar{V}_1$ and $\bar{V}_2$ the method of intercepts is convenient. suppose we have mixtures where to the total number of moles is one. For such systems, $n_1 + n_2 = 1$

so, $dn_1 + dn_2 = 0$, Also, n_1 and n_2 are equal to the respective mole-fractions, N_1 and N_2

i.e. $n_1 = N_1$ and $n_2 = N_2$. Since $V = f(n_1, n_2)$ by differentiation, $dV = \left(\frac{\delta V}{\delta n_1}\right) dn_1 + \frac{\delta V}{\delta n_2} = \bar{V}_1 dn_1 + \bar{V}_2 dn_2$

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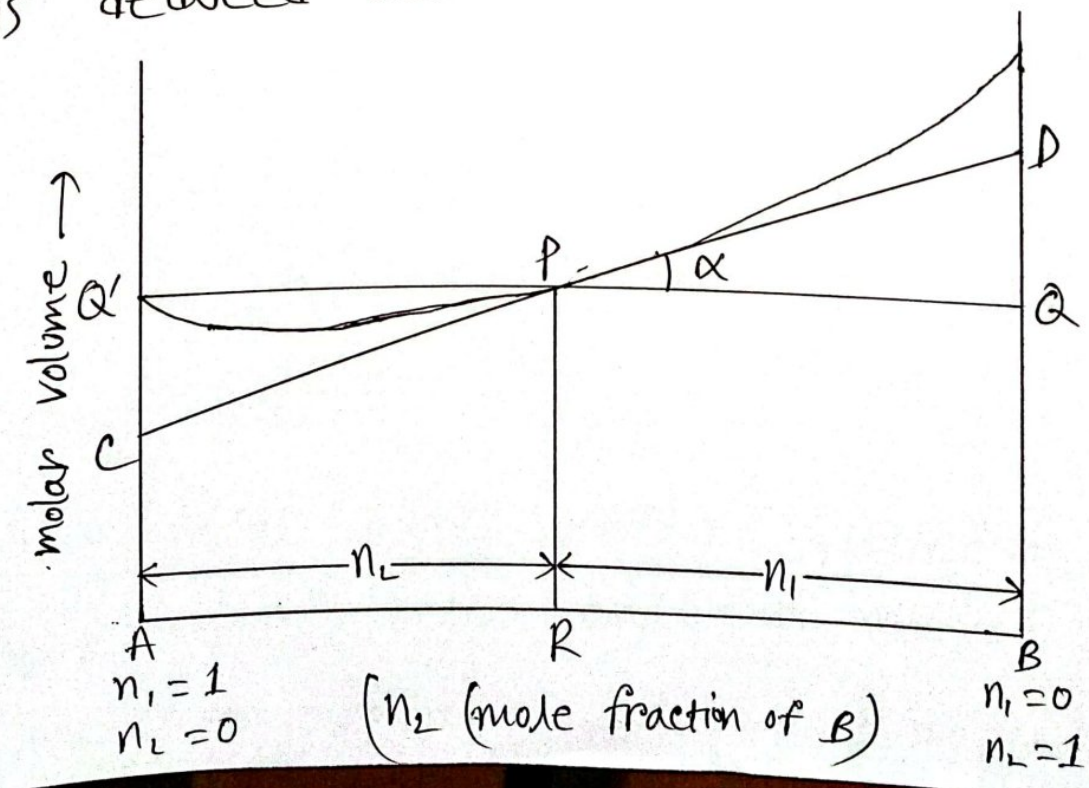
or $dV = \bar{V}_1 dn_1 + \bar{V}_2 dn_2$

$= dn_2 (\bar{V}_2 - \bar{V}_1)$

$\therefore \left(\frac{dV}{dn_2}\right) = \left(\frac{dV}{dN_2}\right) = \bar{V}_2 - \bar{V}_1$ ----(5)

This means that if the total volume of a system of two component is plotted against mole-fraction, the intercept, AC on the $n_1 = 1$ axis is equat to $\bar{V}_1$ and intercept BD on the $n_2 = 1$ axis is equal to $\bar{V}_2$. This is deduced below:



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the slop of the tangent is

$$\tan\alpha = \left(\frac{dV}{dn_2}\right) = \bar{V}_2 - \bar{V}_1$$

Now,

$$BD = BQ + QD = PR + PQ\tan\alpha$$

$$= V + n_1\tan\alpha$$

$$= \bar{V}_1 n_1 + \bar{V}_2 n_2 + n_1(\bar{V}_2 - \bar{V}_1)$$

$$= \bar{V}_2(n_1 + n_2) = \bar{V}_2$$

Similarly,

$$AC = AQ' - CQ' = RP - PQ'\tan\alpha$$

$$= V - n_1\tan\alpha$$

$$= \bar{V}_1 n_1 + \bar{V}_2 n_2 - n_2(\bar{V}_2 - \bar{V}_1)$$

$$= \bar{V}_1(n_1 + n_2)$$

$$= \bar{V}_1$$

The molal volume could be plotted against the mole fraction and the method of intercept applied to get the partial molal volume

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Apparatus and chemical required:

1. westphal balance.
2. Methanol
3. Distilled water

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Group: 09

DATA:

Data table for the determination of weight lost at different concentration of alcohol solution.

% of alcohol	description of weight	weight lost (g)
0%	$(0.6\times5)+(0.5\times1)+(0.4\times0.5)$	3.70
15%	$(0.6\times5)+(0.5\times1)+(0.3\times0.5)$	3.65
30%	$(0.6\times5)+(0.5\times1)+(0.2\times0.5)+(0.4\times0.05)$	3.62
45%	$(0.6\times5)+(0.5\times1)+(0.1\times0.5)+(0.2\times0.05)$	3.56
60%	$(0.6\times5)+(0.3\times1)+(0.2\times0.5)+(0.4\times0.05)$	3.42
75%	$(0.6\times5)+(0.2\times1)+(0.1\times0.5)+(0.9\times0.05)$	3.27
90%	$(0.6\times5)+(0.1\times1)+(0.9\times0.05)$	3.15
100%	$(0.6\times5)+(0.1\times0.5)+(0.2\times0.05)$	3.06

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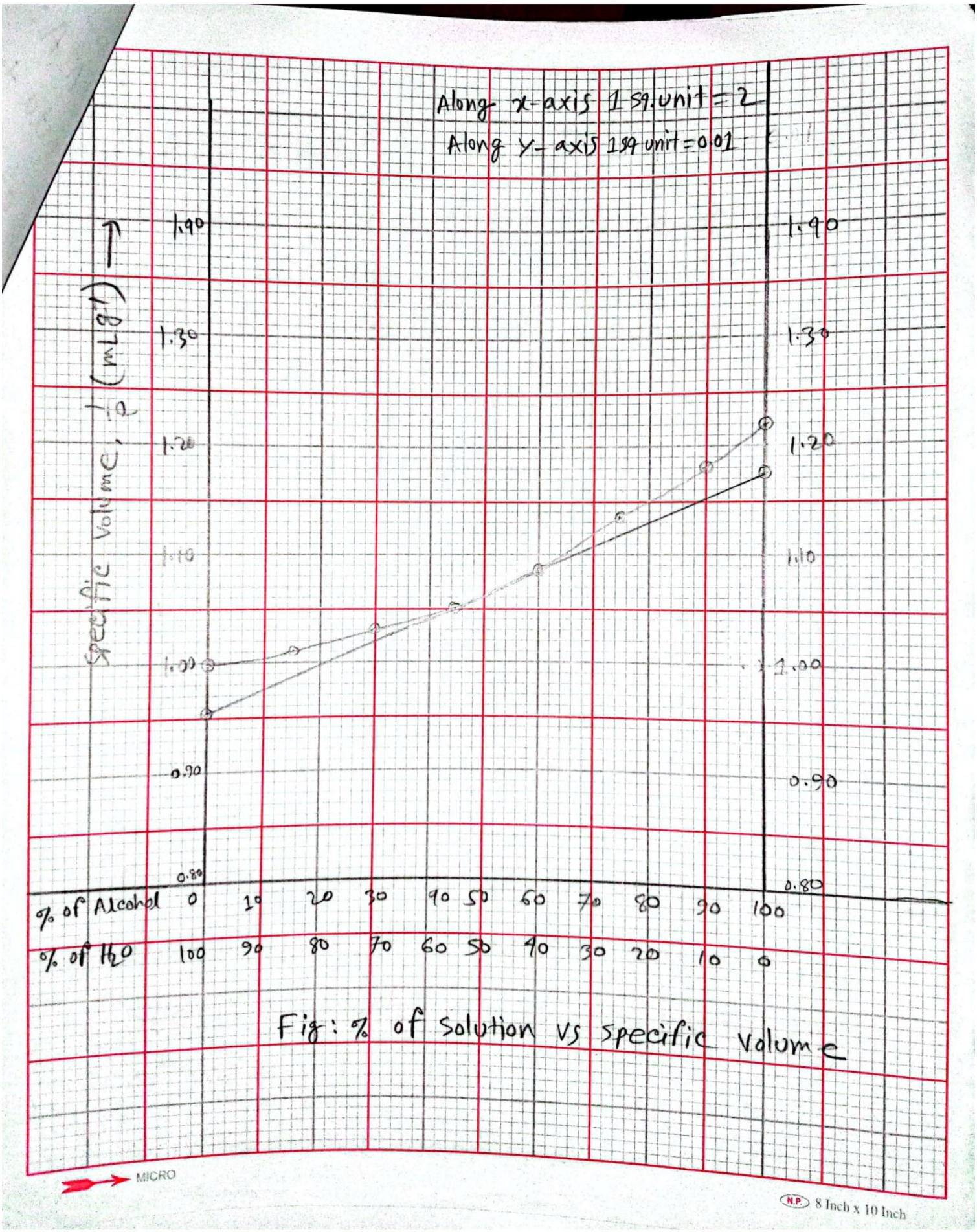

Fig: % of solution vs specific volume

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CALCULATION:

Calculation for specific volume of alcohol & water mixture.

weight loss in water (gm)	% of alcohol in mixture	weight of alcohol + water	Density $\rho = \frac{W_{alc}}{W_{H_2O}}$	specific volume $\frac{1}{\rho}$ (mL/g)
3.70	0	3.70	1.00	1.00
3.70	15	3.65	0.98	1.01
3.70	30	3.62	0.97	1.03
3.70	45	3.56	0.96	1.05
3.70	60	3.42	0.92	1.09
3.70	75	3.27	0.88	1.14
3.70	90	3.15	0.85	1.18
3.70	100	3.06	0.82	1.22

From the specific volume vs percentage of alcohol plot, it is notice that partial specific volume of H_2O = 0.96 mlg^{-1} partial s. Volume of methanol = 1.18 mL/g

Hence, partial molar volume of $H_2O = \frac{1}{\rho} \times M$

$= 0.96 \times 18$

$= 17.28$ mL/mol

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Partial molar volume of methanol $= \frac{1}{\rho} \times M$

$= 1.18 \times 32$

$= 37.76$ mL/mol

RESULT: The partial molar volume of water is 17.28 mL/mol and methanol is 37.76 mL/mol in ale-water system.

DISCUSSION:

1. The loss of weight increases as the percentage of water increases in mixture.
2. This experiment should be carried out in air protected laboratory to obtain accurate result.
3. Apparatus was calibrated by measuring solution.
4. The loss of water and specific volume does not vary linearly.
5. The measurement was taken carefully.
6. The error may reduce by air protected balance.
7. The loss of weight decreases as the increase of percentage of methanol due to low density.

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Jahangirnagar University
Department of Chemistry

Lab Report

Course Title: Physical Chemistry Practical (III)

Course No: CHEM - 315

Experiment No: 05

Experiment Name: Determination of the partial molal volume of solutions (Alcohol-Water system) by Pycnometer.

Submitted To

Dr. Mohammed Abdullah Khan

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Session: 2020-2021

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Experiment NO: 5

Name of Experiment: Determination the partial molal volume of solutions (alcohol-water system) by pyenometer.

THEORY: The partial molal property of a component of a solution may be defined as follows: Let, G represent any extensive property of binary solution; at constant temperature and pressure, The partial molal quantity of component 1 is defined by the relation -

$$\bar{G}_1 = \left(\frac{\partial G}{\partial n_1}\right)_{n_2, T, P} \quad ---(1)$$

Similarly for component 2,

$$\bar{G}_2 = \left(\frac{\partial G}{\partial n_2}\right)_{n_1, T, P} \quad ---(11)$$

Since any extensive property G of a

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mixture is a homogeneous function of degree one of the two independent variables (n_1 and n_2), it follows from Euler's theorem that –

$$G_t(n_1 \text{ and } n_2) = n_1\bar{G}_1 + n_2\bar{G}_2 \quad (T, P \text{ constant}) \text{ ---} ③$$

In the case of the volume of the solution by similar resoning we have –

$$V(n_1 \text{ and } n_2) = n_1\bar{V}_1 + n_2\bar{V}_2 \quad (T, P \text{ constant}) \text{ ---} ④$$

For this system, $n_1 + n_2 = 1$.

So, $dn_1 + dn_2 = 0$. also n_1 and n_2 are equal to the N_1 and N_2 i.e. $n_1 = N_1$ and $n_2 = N_2$

Since $V = f(n_1, n_2)$ by differentiation,

$$dV = \left(\frac{\delta V}{\delta n_1}\right) dn_1 + \left(\frac{\delta V}{\delta n_2}\right) dn_2 = \bar{V}_1 dn_1 + \bar{V}_2 dn_2$$

or, $dV = \bar{V}_1 dn_2 + \bar{V}_2 dn_2$

$$= dn_2(\bar{V}_2 - \bar{V}_1)$$

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$$\therefore \left(\frac{dV}{dn_2}\right) = \left(\frac{dV}{dN_2}\right) = \bar{V}_2 - \bar{V}_1 \quad - - - (5)$$

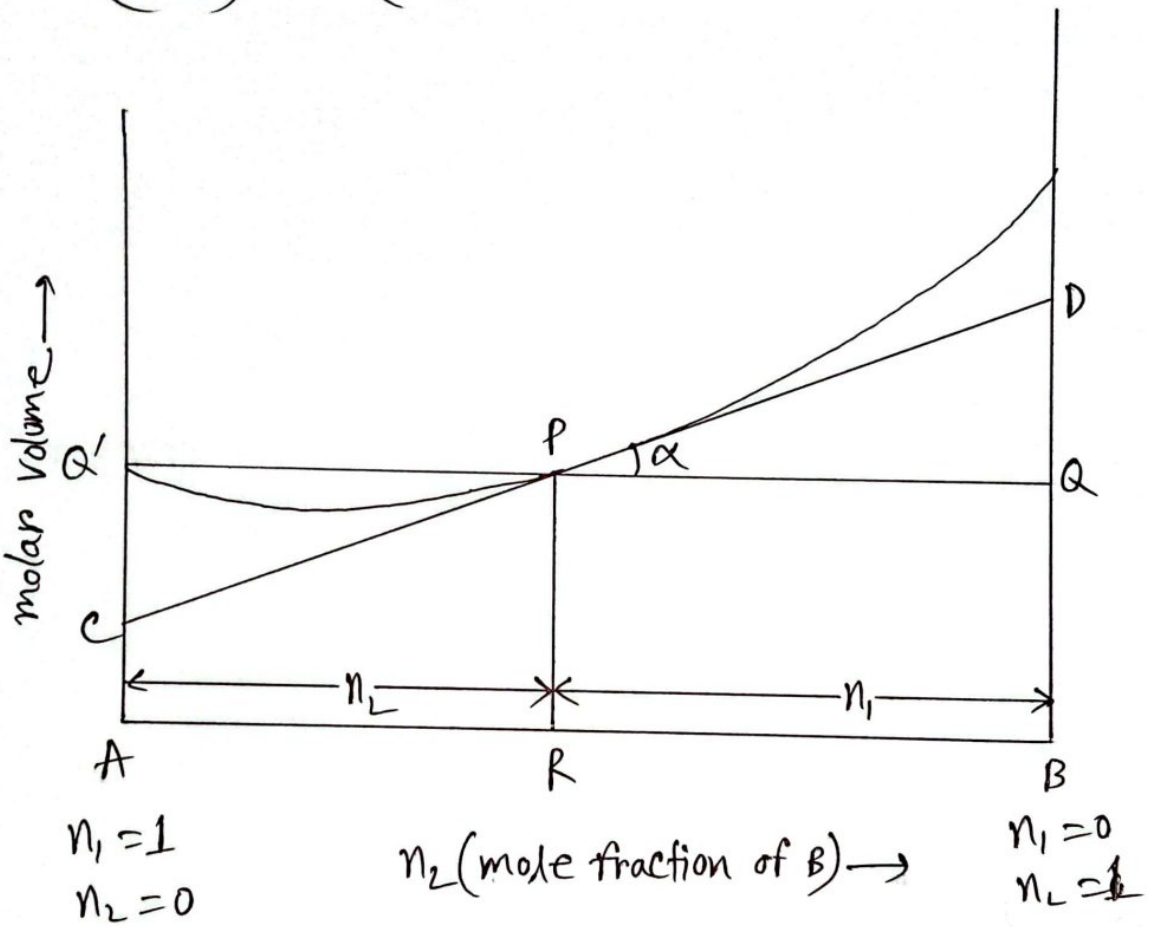

Fig. 1: Molar volume vs mole fraction

Let α be the angle between the tangent and the line V=0. The slope of tangent is,

$$\tan\alpha = \left(\frac{dV}{dn_2}\right) = \bar{V}_2 - \bar{V}_1$$

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Now, $BD = BQ + QD = PR + PQ \tan\alpha$

$= V + n_1 \tan\alpha$

$= \bar{V}_1 n_1 + \bar{V}_2 n_2 + n_1 (\bar{V}_2 - \bar{V}_1)$

$= \bar{V}_2 (n_2 + n_1)$

$= \bar{V}_2$

Similarly, $AC = AQ' - CQ' = PR - PQ' \tan\alpha$

$= V - n_1 \tan\alpha$

$= \bar{V}_1 (n_1 + n_2)$

$= \bar{V}_1$

The molal volume could be plotted against the mole fraction and the method of intercept applied to get the partial molal volume.

Apparatus and chemical required:

1. pycnometer
2. ~~alchol~~ alcohol.
3. distilled water

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Experiment NO: 05

Experiment Name: Determination of the partial molar volume of solution (alcohol-water) by pyenometer.

DATA:

1. Mass of the dry pyenometer = 15.48 gm
2. Mass of the pyenometer with water = 25.65gm
3. Mass of the pyenometer with 15% methanol = 25.41gm
4. Mass of the pyenometer with 30% methanol = 25.19gm
5. Mass of the pyenometer with 45% methanol = 24.86gm
6. Mass of the pyenometer with 60% methanol = 24.49 gm
7. Mass of the pyenometer with 75% methanol = 24.14gm
8. Mass of the pyenometer with 90% methanol = 23.72gm
9. Mass of the pyenometer with 100% methanol = 23.46gm
10. Mass of water, $m_{H_2O} = (25.65 - 15.48)$ gm

$= 10.17$ gm

29.5.24

CALCULATION:

Table-1: Data for solution of A (water) and B(alcohol).

Aqueous methanol solution	Wt. of methanol W_B (g)	No of mole(CH_3COOH) $n_B = \frac{W_B}{M_B}$	Wt. of water W_A (g)	No of mole(H_2O) $n_A = \frac{W_A}{M_A}$	Mole fraction of methanol $X_B = \frac{n_B}{n_A + n_B}$	mole fraction of water, $X_A = 1 - X_B$
water	0	0.00	100	5.56	0.00	1.00
15%	15	0.47	85	4.72	0.09	0.91
30%	30	0.94	70	3.89	0.19	0.81
45%	45	1.41	55	3.06	0.32	0.68
60%	60	1.88	40	2.22	0.46	0.54
75%	75	2.34	25	1.39	0.63	0.37
90%	90	2.81	10	0.56	0.83	0.17
100%	100	3.13	0	0.00	1.00	0.00

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Table-2

Data table for the determination of the molar volume of alcohol (B) and water (A):

aqueous solution	For CH_3COOH $X_B = 1 - X_A$	For H_2O X_A	For solution $M_{av} = X_A M_A + (1-X_A) M_B$ g mol^{-1}	Mass of pycnometer + liquid (g)	Mass of liquid, m_L (g)	For solution Density, $d_L = \frac{m_L}{m_{H_2O}} d_{H_2O}$ (g cm^{-3})	Molar volume $V_m = \frac{M_{av}}{d_L}$ (cm^3/mol)
water	0.00	1.00	18.00	25.65	10.17	1.00	18.00
15% MeOH	0.09	0.91	19.26	25.41	9.93	0.98	19.65
30% MeOH	0.19	0.81	20.66	25.19	9.71	0.95	21.75
45% MeOH	0.32	0.68	22.48	24.86	9.38	0.92	24.43
60% MeOH	0.46	0.59	24.44	24.49	9.01	0.89	27.51
75% MeOH	0.63	0.37	26.82	24.19	8.66	0.85	31.55
90% MeOH	0.83	0.17	29.62	23.72	8.29	0.81	36.57
100% MeOH	1.00	0.00	32.00	23.46	7.98	0.78	41.03

From the graph, the partial molar volume of water is 18 mL mol^{-1} and the partial molar volume of methanol is 41.03 mL mol^{-1}.

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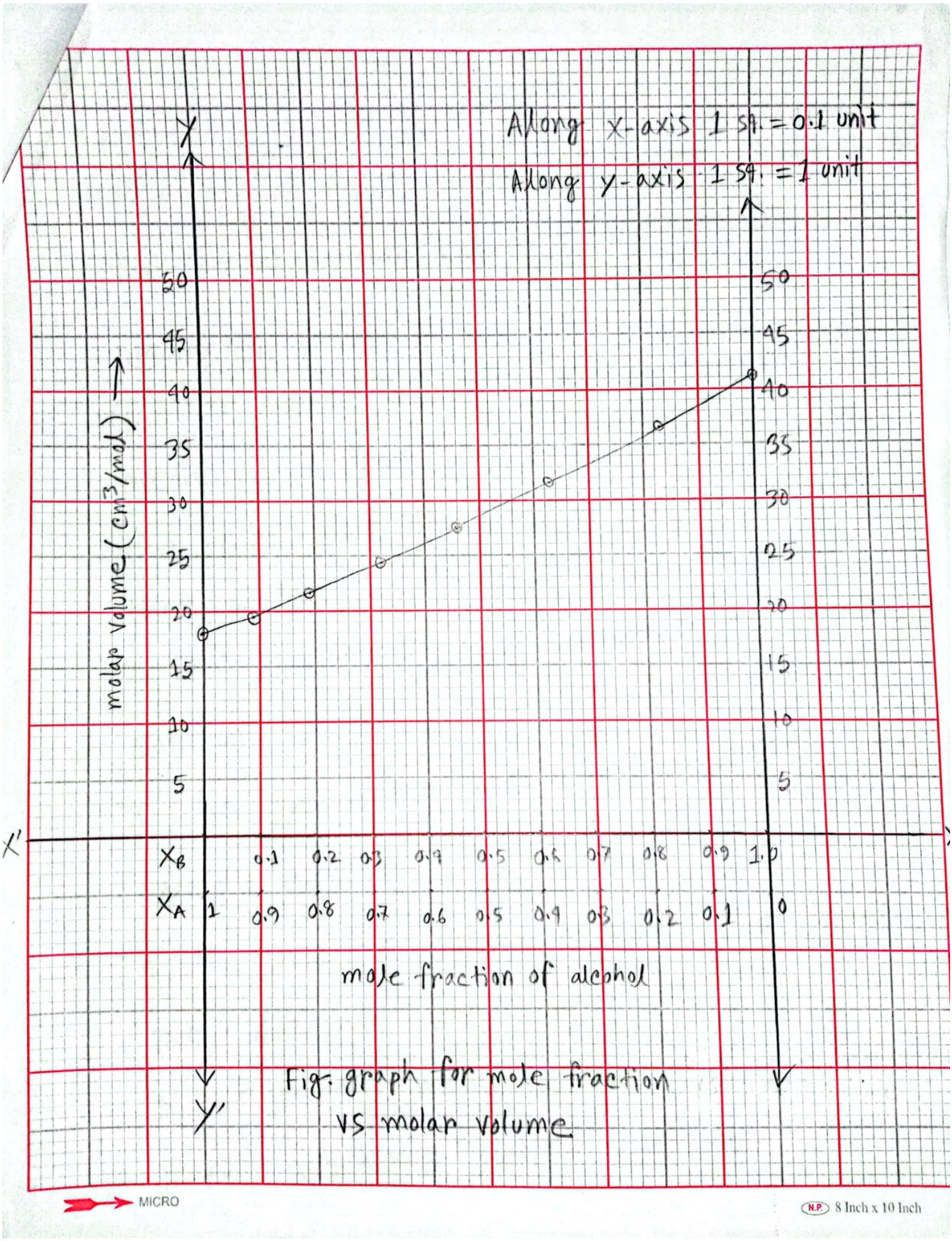

Fig: graph for mole fraction vs molar volume

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RESULT:

1. The partial molar volume of water is 18 mL mol^{-1}.

2. The partial molar volume of methanol is 41.03 mL mol^{-1}.

DISCUSSION:

① Molar volume increases as percentage of methanol increases in solution, this is due to the fact that, the alcohol is less denser than water.

2. The pycnometer was calibrated by measuring solution.

3. The pycnometer was filled accurately on weight carefully.

4. The specific volume of solution of alchol and H_2O does not vary linearly.

5. The partial molar volume of was found from graph which was plotted carefully.

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Jahangirnagar University
Department of Chemistry

Lab Report

Course Title: Physical Chemistry Practical (III)

Course No: CHEM - 315

Experiment No: 06

Experiment Name: Finding (a) the specific rotation of sucrose (b) concentration of an unknown sucrose solution (c) influence of solvent on the rotation of sucrose solution by polarimetric method.

Submitted To

Dr. Mohammed Abdullah Khan

Professor

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Submitted By:

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Group No: 09

Session: 2020-2021

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Experiment No: 06

Name of experiment: Finding (a) the specific rotation of sucrose (b) concentration of an unknown sucrose solution (c) Influence of solvent on the rotation of sucrose solution by polarimetric method.

Theory: A solution of sucrose is optically active, it has ability to rotate the plane of polarization. Fructose is leavo rotatory, which means that it can rotate the plane of polarization. Glucose is dextro rotatory, and the mixture of products has the resultant leavo-rotatory.

The angle of rotation depends on:

(a) Temperature (b) wavelength of light used (c) length of the liquid column. (d) nature of the substrate

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The specific rotation of a substance is given by:

$$[\alpha]_\lambda = 100\ \theta/cl$$

Apparatus and chemical required:

1. polarimeter
2. 100ml volumetric flask
3. pipette
4. sucrose
5. 2N HCl.

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Experiment NO: 06

Experiment Name: Finding out (a) the specific rotation of sucrose (b) concentration of an unknown sucrose solution (c) Influence of solvent on the rotation of sucrose solution by polarimetric method.

DATA:

Measurement of angle of rotation (θ) of the Sucrose solution by polarimetric method.

% of sucrose solution	circular scale rotation (cm)	vernier scale readings (cm)	vernier constant (VC)	Total scale reading (cm)
5%	5	7	0.05	5.40
10%	12	7	0.05	12.35
15%	19	4	0.05	19.30
20%	26	2	0.05	26.10

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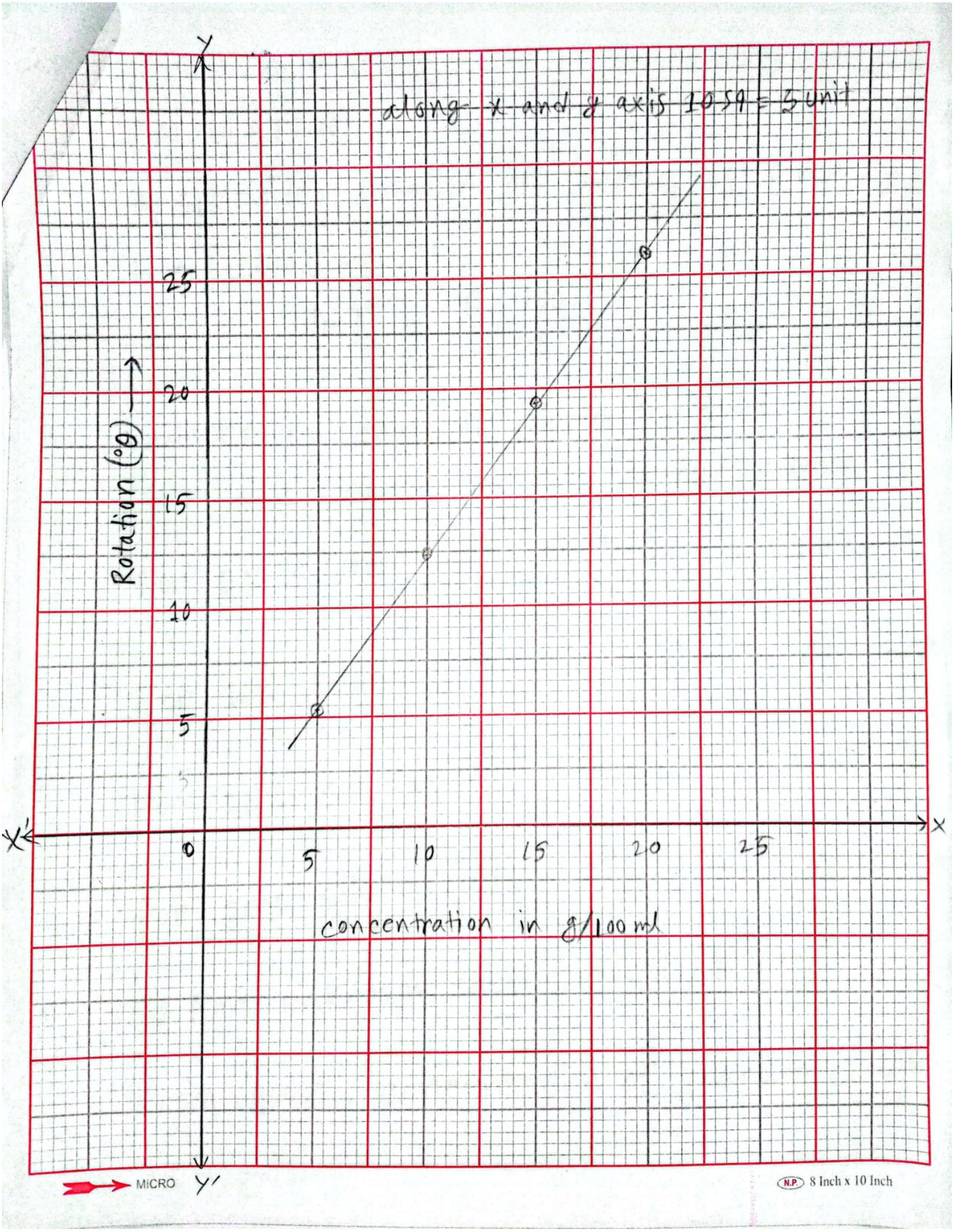

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CALCULATION:

(a) specific rotation of sucrose is given by the equation,

$$[\alpha]_\lambda^t = \frac{\theta \times 100}{l \times c}$$

where, l = length of polarimeter tube =

c = concentration of sucrose solution

For 5% solution, $[\alpha]_\lambda^t = \frac{5.40 \times 100}{2 \times 5} = 54.0^\circ cm^2 g^{-1}$

For 10% solution $[\alpha]_\lambda^t = \frac{12.35 \times 100}{2 \times 10} = 61.75^\circ cm^2 g^{-1}$

For 15% solution $[\alpha]_\lambda^t = \frac{19.30 \times 100}{2 \times 15} = 64.33^\circ cm^2 g^{-1}$

For 20% solution $[\alpha]_\lambda^t = \frac{26.10 \times 100}{2 \times 20} = 65.25^\circ cm^2 g^{-1}$

(b) For unknown solution found $\theta = 8^\circ$

From graph, concentration of unknown solution is 13gm/100ml.

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RESULT:

(a) The specific rotation of 5%, 10%, 15%, ~~20%~~ and 20% solutions are $54.0° cm^2 g^{-1}$, $61.75° cm^2 g^{-1}$, $64.33° cm^2 g^{-1}$ and $65.25° cm^2 g^{-1}$ respectively.

(b) concentration of unknown solution is 13gm/100ml.

(c) The influence of solvent is the lowering of the rotation of the sucrose solution.

DISCUSSION:

1. From this experiment, we find out the specific rotation of the sucrose solution is +65.135°. That means the sucrose solution is optically active and dextrorotatory.

2. I cleaned the polarimeter tube carefully.

3. In this experiment, the used water was

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dust free.

4. Whenever a solution is changed, I ringed the tube with the new solution.

5. There was no air bubble inside the tube.

6. The position of analyser was set accurately.

7. The temperature and wavelength was fixed.

8. The sucrose solution were prepared carefully.

9. Analyser was rotated until the best match is obtained between the dark and bright parts of the field view.

10. The weight and readings was taken carefully.

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Jahangirnagar University
Department of Chemistry

Lab Report

Course Title: Physical Chemistry Practical (III)

Course No: CHEM - 315

Experiment No: 07

Experiment Name: Determination of (a) the absorption curve of coloured substances in solution, (b) the validity of Beer-lambert law, and (c) the strength of an unknown solution.

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Session: 2020-2021

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Experiment NO: 07

Name of Experiment: Determination (a) the absorption curve of coloured substances in solution, (b) the validity of Beer-Lambert law, and (c) the strength of an unknown solution.

Theory: Absorption spectroscopy can be used to quantify the absorbing species present in the sample. According to the beer-lambert's law, decrease in intensity ($-dI$) is proportional to the thickness of the medium (dl) and intensity of incident light (I). In such cases decrease in intensity is also proportional to molar concentration (c). So,

$$-dI = K.I.dl.c \quad (K \text{ is a constant})$$

Beer-lambert's law states that for a solution,

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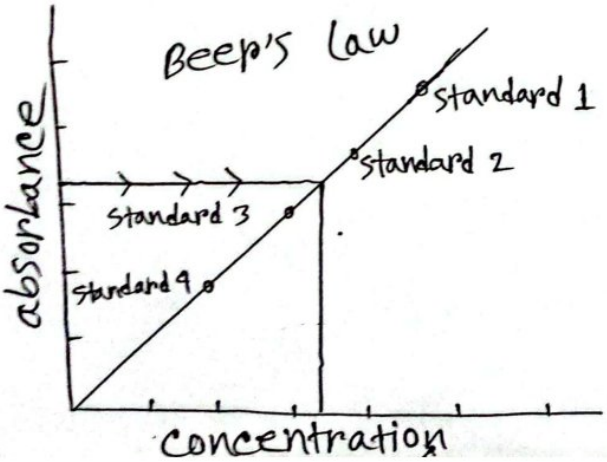
the absorbance of the sample (A) is given by: $A = \log(I_0/I) = \epsilon c l = O.D.$

Since ϵ depends on wavelength and nature of the substance so for a given substance at a given frequency and in a given photochemical cell O.D. = costant * C.

Hence, a plot of O.D. vs C will be a straight line through origin. This is Beer's law.

Absorbance has no units. If 'I' is expressed in cm, and 'c' in mol dm^{-3}, then ϵ has the units of $mol^{-1} dm^3 cm^{-1}$. A linear plot of absorbance vs concentration through origin would verify the Beer-Lambert's law as shown below:



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Apparatus and chemical required:

1. spectronic 20
2. Five 100ml volumetric flask
3. $KMnO_4$

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Experiment NO: 07

Name of the experiment: Determination of the absorption curve of coloured substances in solution and the validity of Beer-lambert law.

Data for the determination of λ max value for 15% of given 0.001M $KMnO_4$ colored substances.

wavelength λ(nm)	absorbance	wavelength λ(nm)	absorbance
480	0.115	530	0.356
490	0.170	590	0.339
500	0.239	550	0.317
510	0.285	560	0.215
520	0.341	570	0.190

Data for the determination of absorbance of different concentration of given coloured substance at the λ_{max}.

wavelength λ_{max} (nm)	conc. of $KMnO_4$ (M)	absorbance
530	0.0005	0.130
530	0.00010	0.250
530	0.00015	0.377
530	0.00020	0.511
530	0.00025	0.616

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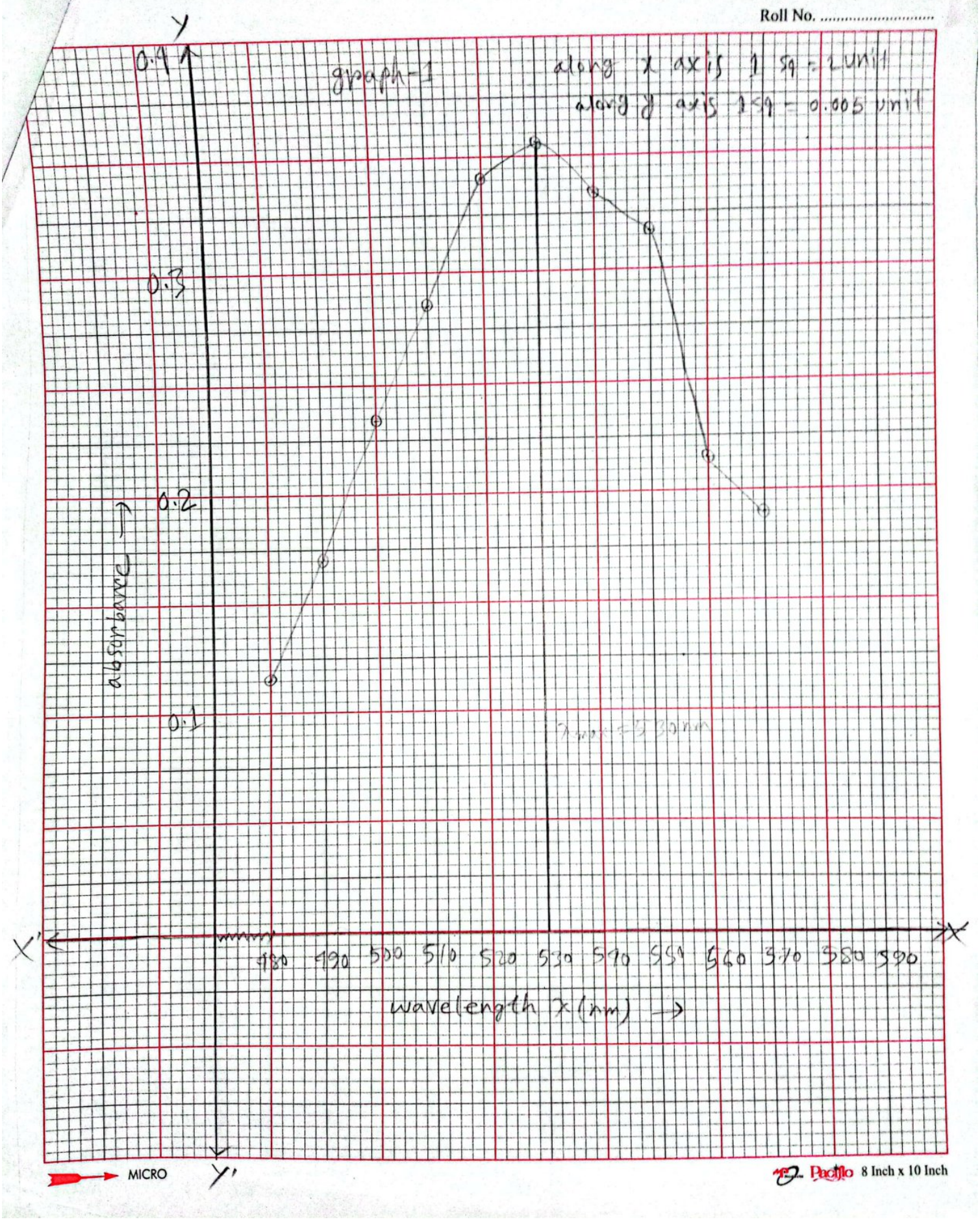

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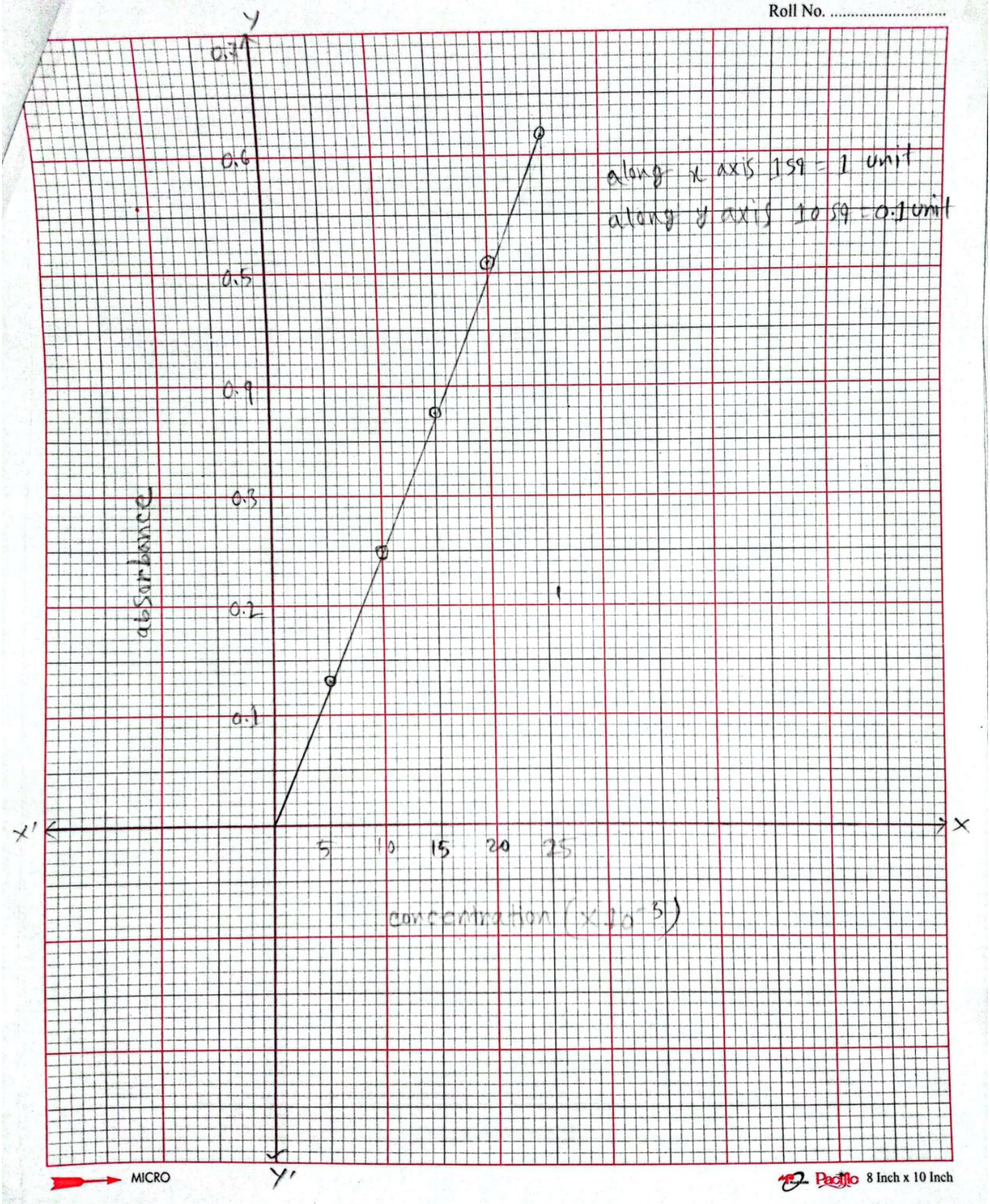

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CALCULATION:

1. From graph-1 absorption curve, λ_{max} = 590 nm.

2. Beer lambert law is verified because the straight line pass through the origin.

3. From the plot of absorbance Vs concentration the slope is = $\frac{y_2 - y_1}{x_2 - x_1}$

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RESULT:

1. From the graph, $\lambda_{max} = 530$ nm
2. ~~Gpha~~ Graph-2 show the validity of Beer-lambert law.

DISCUSSION:

1. All the solution was prepared accurately.
2. λ_{max} was determined graphically for accurate value.
3. while determining λ_{max} for $KMnO_4$, water was used as a blank.
4. The solvent was filled fully and cleaned carefully.
5. All the measurement and graph was plotted carefully.
6. There were not ~~parent~~ present any air bubble in the tube containing $KMnO_4$ solution.
7. In this experimen micromatic light is used.
8. There may make some error in this experiment.

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Jahangirnagar University
Department of Chemistry

Lab Report

Course Title: Physical Chemistry Practical (III)

Course No: CHEM - 315

Experiment No: 08

Experiment Name: Determination of the specific reaction rate of base catalysed hydrolysis of an ester by conductometric method.

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Session: 2020-2021

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Experiment NO: 08

Name of experiment: determination the specific reaction rate of base catalysed hydrolysis of an ester by conductometric method.

THEORY: In the presence of alkali ethyl acetate undergoes hydrolysis in accordance with

$$CH_3COOC_2H_5 + OH^- = CH_3COO^- + C_2H_5OH$$

The rate of saponification is directly proportional to both concentrations of ester and alkali, and the reaction therefore, is a second order one, i.e.

$$K = \frac{1}{t}\frac{x}{a(a-x)} \; ; \; C_t = \frac{1}{aK}\frac{C_0 - C_t}{t} + C_\infty$$

where a and b are the initial concentration of ester and alkali, respectively ($a=b$ in this case) and K is the reaction rate constant.

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Apparatus and chemical required:

1. conductometer and cell
2. thermostat bath
3. 250 ml conical flask
4. pipette 0.1M NaOH
5. 0.1M Ethyl acetate or 0.1M Methyl acetate.

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Date: 15.05.2024
Roll: 200
Group: 09

Experiment No: 08

Name of experiment: Determination of the specific reaction rate of base catalysed hydrolysis of an ester by conductometric method.

Data:

Table for the measurement of the conductance.

Time (min)	conductance ($ms\,cm^{-1}$)	Time (min)	conductance ($ms\,cm^{-1}$)
0.5	12.30	6.5	7.40
1.0	11.10	7.0	7.30
1.5	10.20	7.5	7.21
2.0	9.65	8.0	7.10
2.5	9.10	8.5	7.04
3.0	8.70	9.0	6.90
3.5	8.40	9.5	6.85
4.0	8.15	10.0	6.83
4.5	8.10	10.5	6.75
5.0	7.83	11.0	6.69
5.5	7.63	11.5	6.65
6.0	7.50	12.0	6.58

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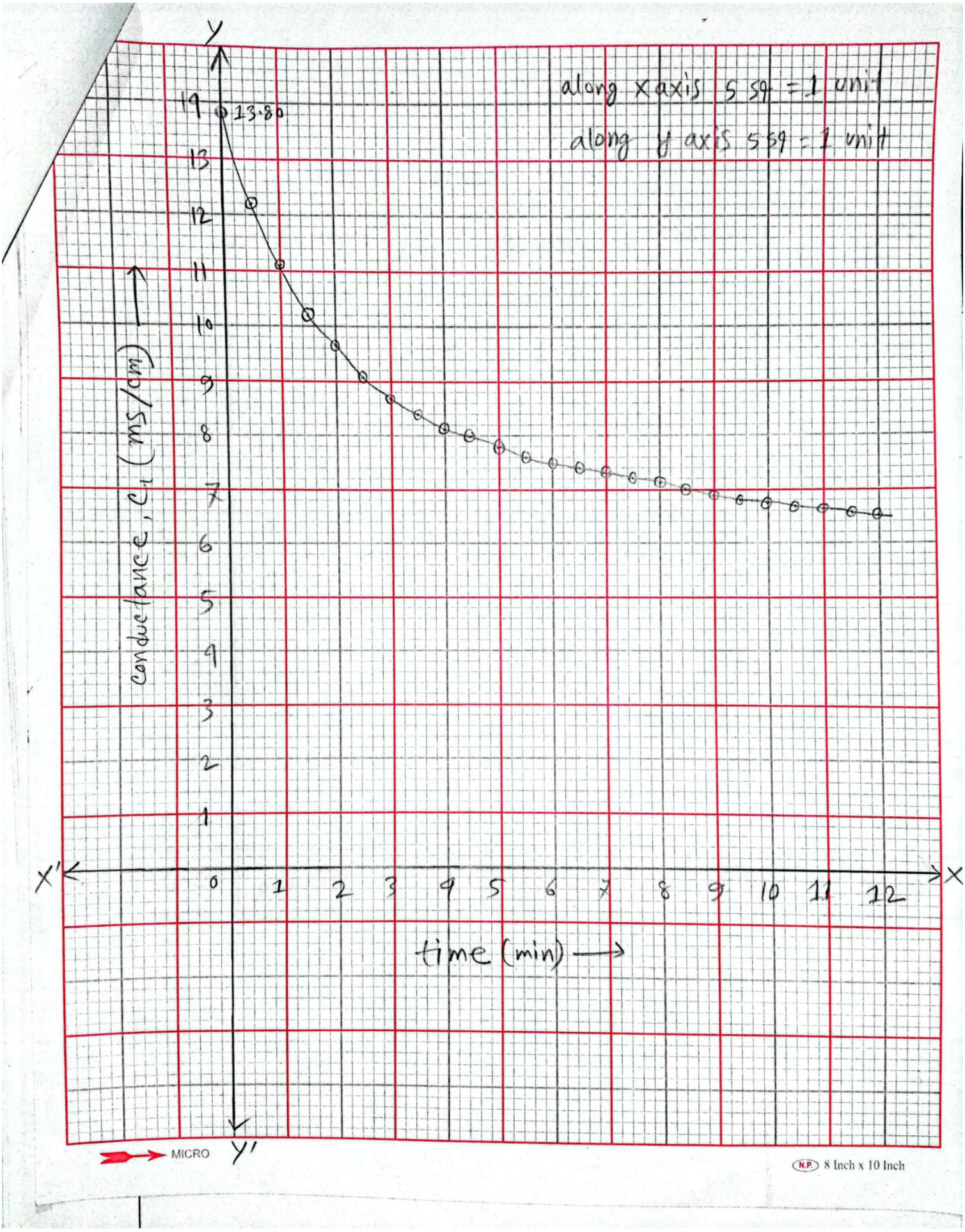

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CALCULATION:

From the first graph, $C_o = 13.8$ ms/cm

Time, t(min)	conductance C_t (ms/cm)	$\frac{C_o - C_t}{t}$	Time, t(min)	conductance C_t (ms/cm)	$\frac{C_o - C_t}{t}$
0.5	12.30	3.0	6.5	7.40	0.98
1.0	11.10	2.7	7.0	7.30	0.93
1.5	10.20	2.4	7.5	7.21	0.89
2.0	9.65	2.07	8.0	7.10	0.89
2.5	9.10	1.88	8.5	7.09	0.79
3.0	8.70	1.70	9.0	6.90	0.76
3.5	8.90	1.59	9.5	6.85	0.73
4.0	8.15	1.91	10.0	6.83	0.69
4.5	8.10	1.27	10.5	6.75	0.67
5.0	7.83	1.19	11.0	6.69	0.69
5.5	7.63	1.12	11.5	6.65	0.62
6.0	7.50	1.05	12.0	6.58	0.60

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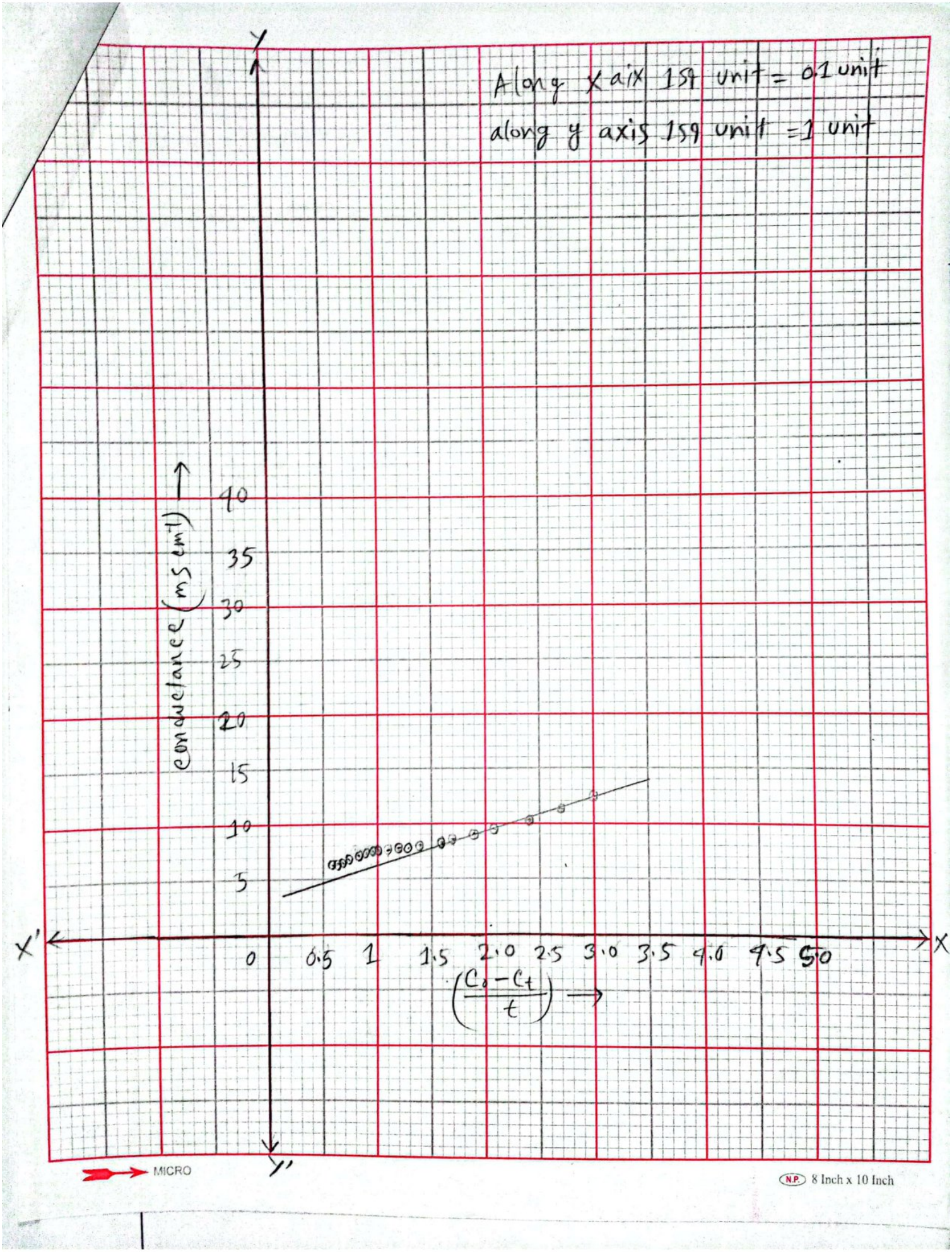

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From graph-2

$$\text{The slope} = \frac{11.10 - 7.30}{2.7 - 0.93}$$

$$= 2.15$$

$\therefore$ Initial concentration of ester is $0.1M = a$

$$\therefore \text{slope} = \frac{1}{aK}$$

$$\Rightarrow K = \frac{1}{a \times \text{slope}}$$

$$= \frac{1}{0.1 \times 2.15}$$

$$= 4.65$$

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RESULT: specific reaction rate of base catalysed hydrolysis of ester is such as ethyl acetate is 4.65.

DISCUSSION:

1. At first I cleaned all the apparatus.
2. The initial concentration and initial conductance of the mixture, was measured carefully from the graph.
3. The conductance of the reaction mixture decreased with increasing of time.
4. All the reading were taken considering as such as care and plotted at graph carefully.
5. Digital conductometer was used to take correct point.
6. There may be some instrumental error.

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Jahangirnagar University
Department of Chemistry

Lab Report

Course Title: Physical Chemistry Practical (III)

Course No: CHEM - 315

Experiment No: 09

Experiment Name: Finding out the equivalent conductance at infinite dilution and the dissociation constant of a weak electrolyte.

Submitted To

Dr. Mohammed Abdullah Khan

Professor

Department of Chemistry

Jahangirnagar University

Savar, Dhaka-1342

Submitted By:

Name: Md Rajib Mia

Roll: 200

Group No: 09

Session: 2020-2021

Department of Chemistry

Jahangirnagar University

Savar, Dhaka-1342

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Experiment No: 09

Name of experiment: Finding out the equivalent conductance at infinite dilution and the dissociation constant of a weak electrolyte.

THEORY: Weak electrolytes dissociate as follows

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

Degree of dissociation, $\alpha = \frac{\Lambda_c}{\Lambda_\infty}$

According to ostwald's dilution law

$$K_c = \frac{\alpha^2 c}{1-\alpha}$$

$$= \frac{\Lambda_c^2 \cdot c}{\Lambda_\infty(\Lambda_\infty - \Lambda_c)}$$

$$\frac{\Lambda_\infty}{\Lambda_c^2} = K_c \frac{(1 - \Lambda_c/\Lambda_\infty)}{\Lambda_c \cdot c}$$

$$\Lambda_c \cdot c = K_c \left(\frac{\Lambda_\infty^2}{\Lambda_c} - \Lambda_\infty\right)$$

Plot of $\Lambda_c \cdot c$ against $1/\Lambda_c$ will thus give a straight line. At $\Lambda_c \cdot c = 0$, $OX = \frac{1}{\Lambda_\infty}$

that is $\Lambda_\infty = \frac{1}{OX}$

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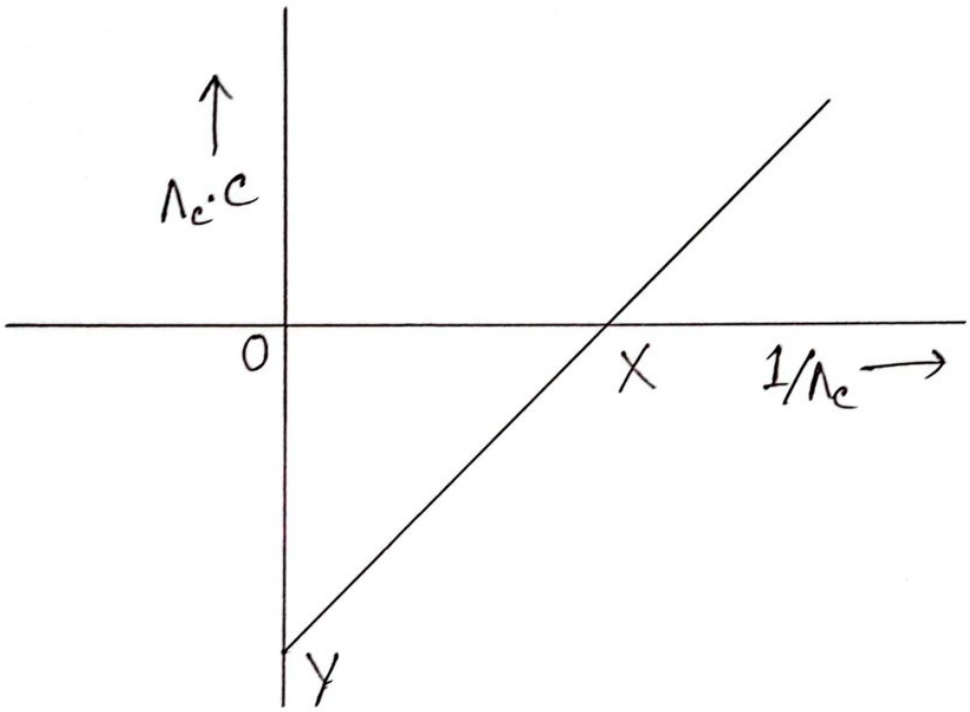
At $\frac{1}{\Lambda_c} = 0$, $OY = -K_c \cdot \Lambda_0$

Therefore $(OX) \cdot (OY) = -K_c$



Apparatus and chemical required:

1. conductometer and cell
2. beaker
3. Stirrer
4. N/10 acetic acid.

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Date: 02.02.2024
Roll: 200
Group: 09

Experiment No: 09

Name of the experiment: Finding out the equivalent conductance at infinite dilution and the dissociation constant of a weak electrolyte.

DATA: Table for the conductance at infinite dilution of electrolyte.

No. of obs.	concentration of CH_3COOH (N)	conductance, Λ_{soln} ($\mu S/cm$)
01	0.0500	431.1
02	0.0333	327.1
03	0.0050	107.1
04	0.0025	72.9
05	0.0010	44.5
06	0.0005	31.7
07	0.0001	22.2
08	Distilled water (H_2O)	9.50

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CALCULATION:

soln conc. C(N)	Specific conductance $K = (\Lambda_{soln} - \Lambda_{H_2O})$ (S/cm)	$\Lambda_c = \frac{1000K}{C}$	$\Lambda_c \cdot C$	$1/\Lambda_c$	$\alpha = \frac{\Lambda_c}{\Lambda_0}$	$K = \frac{\alpha^2 C}{1-\alpha^2}$	$\sqrt{C}$
0.0500	4.21×10^{-4}	8.42	0.421	0.119	0.116	6.8×10^{-4}	0.223
0.0333	3.17×10^{-4}	9.60	0.3168	0.104	0.132	5.8×10^{-4}	0.182
0.0050	9.76×10^{-5}	19.52	0.0976	0.0512	0.269	3.9×10^{-4}	0.0707
0.0025	6.34×10^{-5}	25.36	0.0639	0.0394	0.349	3.9×10^{-4}	0.05
0.0010	3.5×10^{-5}	35.00	0.035	0.0285	0.481	3×10^{-4}	0.032
0.0005	2.22×10^{-5}	44.40	0.0222	0.022	0.611	3.4×10^{-4}	0.022
0.0001	1.27×10^{-5}	124.00	0.0124	0.008	1.709	1.5×10^{-4}	0.01

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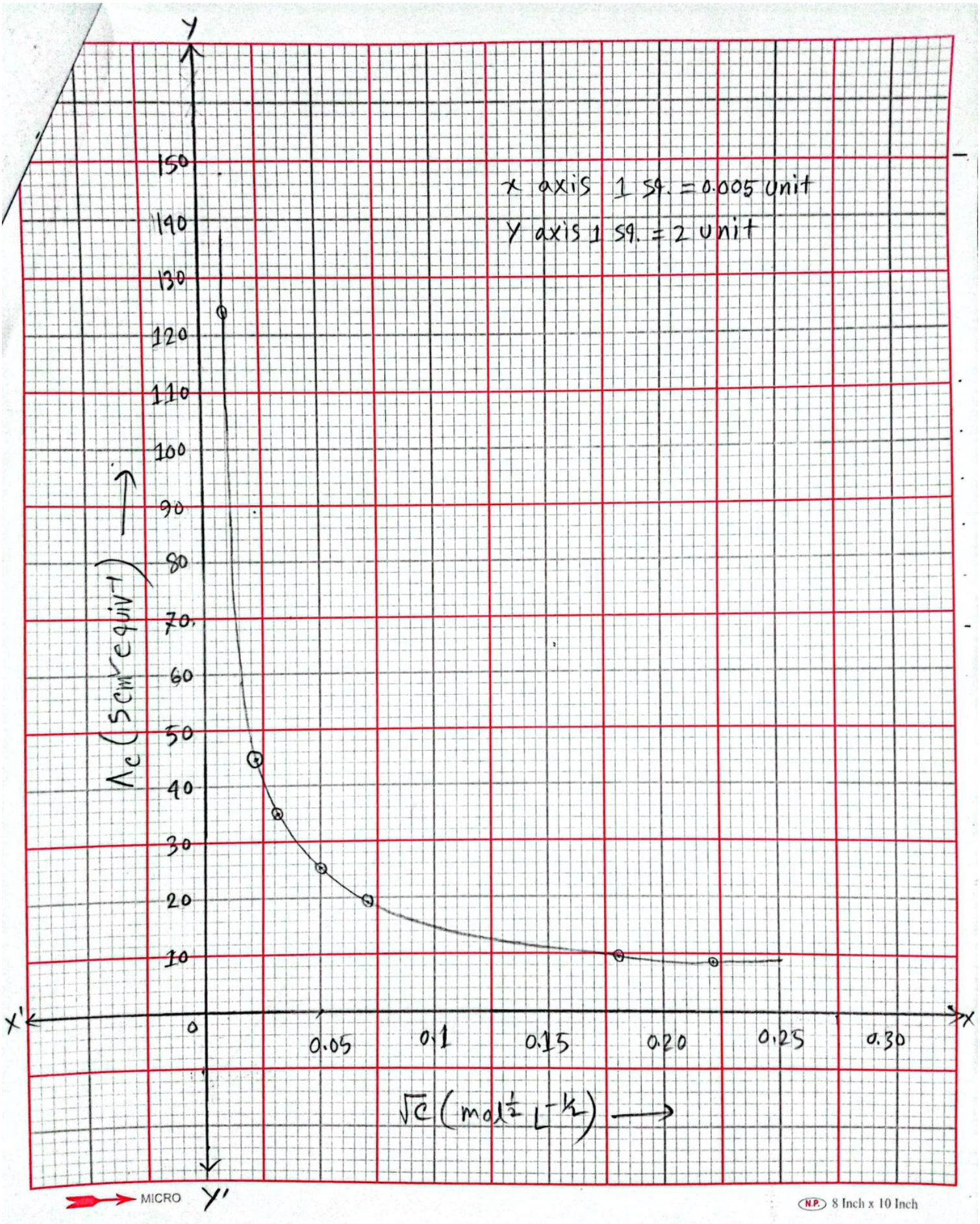

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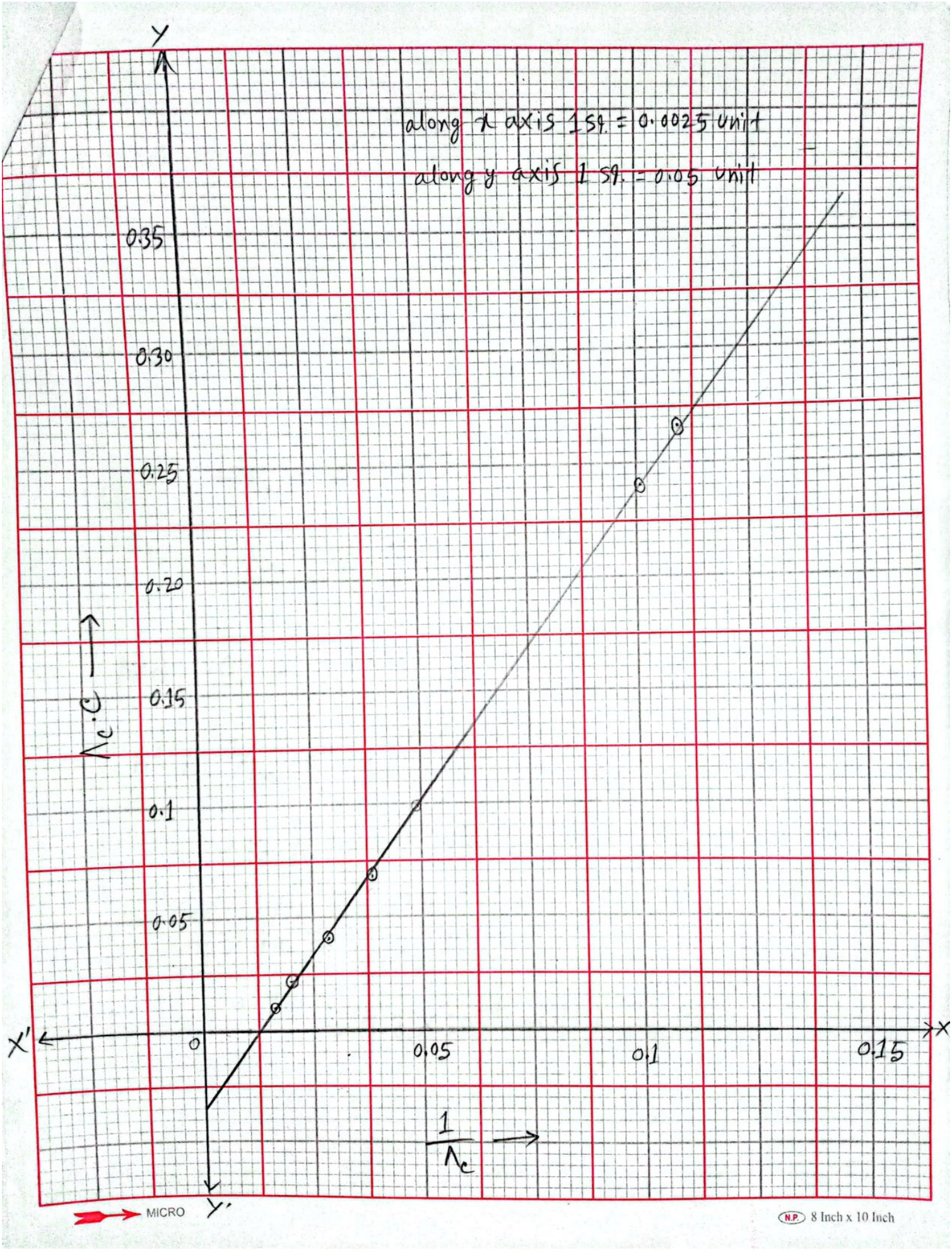

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From graph-2

$$OX_1 = 0.01375$$

at, $\Lambda_c \cdot C = 0$, $OX = \frac{1}{\Lambda_0}$

$\therefore 1/\Lambda_0 = 0.01375$

$\therefore \Lambda_0 = 72.73 \text{ scm}^2 \text{ equiv}^{-1}$

And $OY = -0.035$

At, $1/\Lambda_c = 0$, $OY = -K_c \cdot \Lambda_0$

$\therefore -K_c \cdot \Lambda_0 = -0.035$

$\Rightarrow K_c = \frac{-0.035}{-72.73}$

$= 4.8 \times 10^{-4} \text{ mol L}^{-1}$

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RESULT:

1. The equivalent conductance at infinite dilution of acetic acid is = 72.73 $S\,cm^2\,equiv^{-1}$

2. The dissociation constant of acetic acid is = $4.8 \times 10^{-9}\,mol\,L^{-1}$

Discussion:

1. All the apparatus were cleaned with distilled water after each reading.

2. The equivalent conductance increases due to mobile proton (H^+).

3. conductance of distilled water used in the experiment and subtracted it from the observed conductance of each solution to get specific conductance.

4. All the solutions were prepared accurately.

5. Due to lackings of graph size there had some errors.

6. I plotted the graph very carefully.

7. At dilution increases the conductance is increases due to the degree of dissociation.

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Jahangirnagar University
Department of Chemistry

Lab Report

Course Title: Physical Chemistry Practical (III)

Course No: CHEM - 315

Experiment No: 10

Experiment Name: Conductometric titration of (a) strong and (b) weak acids and drawing their neutralisation curves.

Submitted To

Dr. Mohammed Abdullah Khan
Professor
Department of Chemistry
Jahangirnagar University
Savar, Dhaka-1342

Submitted By:

Name: Md Rajib Mia
Roll: 200
Group No: 09
Session: 2020-2021
Department of Chemistry
Jahangirnagar University
Savar, Dhaka-1342

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Experiment NO: 10

Name of the experiment: conductometric titration of (a) strong and (b) weak acids and to draw their neutralization curves.

THEORY: conductometric titration is the volumetric analysis, based upon the measurement of the conductance during the titration. When one electrolyte is added to another electrolyte. The change in number of free ions causes a change in the conductance.

For example NaOH vs HCl: Before adding NaOH solution to HCl, HCl has high conductivity to highly mobile H^+ ions by OH^- ions of NaOH.

$$H^+Cl^- + Na^+OH^- \longrightarrow Na^+ + Cl^- + H_2O$$

The conductivity of solution decreases until an equivalent point is reached. Further addition

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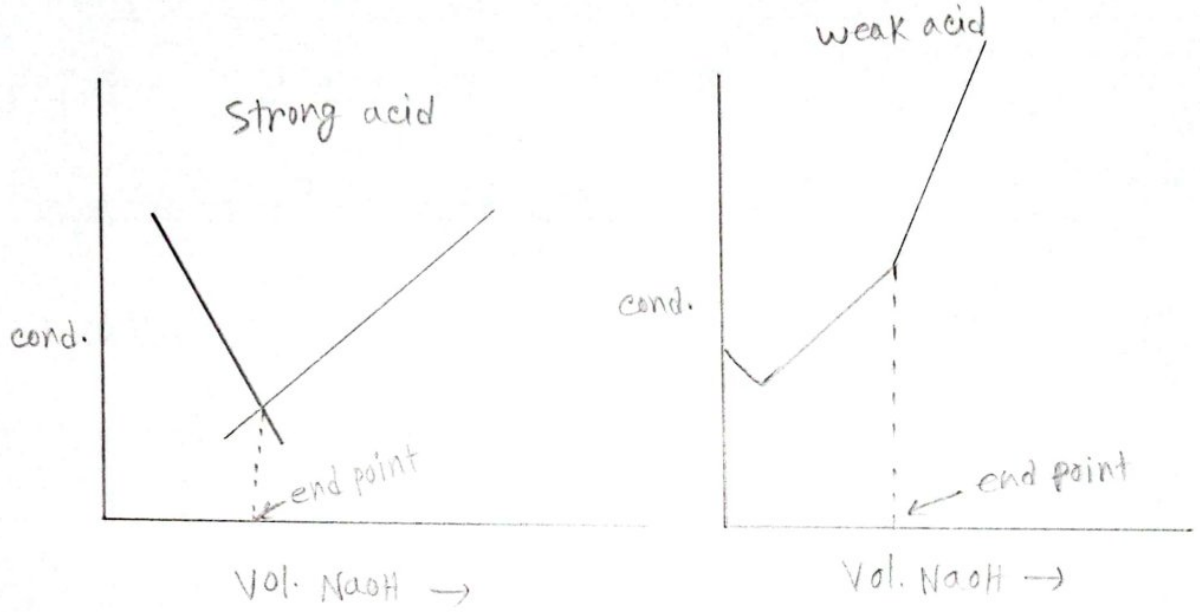
of NaOH increases the conductivity of solution due to increase in the highly mobile OH^- ions.



Weak acid with a strong base, e.g. acetic acid with NaOH: Initially the conductance is low due to feeble ionization of acetic acid. The conductance increases on adding NaOH as NaOH neutralizes the un-dissociated CH_3COOH to CH_3COONa which is the strong electrolyte. The graph near the equivalence point is curved due to hydrolysis of salt CH_3COONa. Beyond the equivalent point

conductance increases more rapidly with the addition of NaOH due to the highly conducting OH^- ions.

Apparatus and chemical required:

1. conductometer and cell
2. beaker
3. volumetric flask
4. HCl
5. Acetic acid
6. NaOH.

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Date : 7.05.2029
Roll : 200
Group : 09

Experiment NO: 10

Name of the experiment : conductometric titration of (a) strong and (b) weak acids and drawing their neutralization curves.

DATA:

Table-a : conductometric titration of strong acid, HCl with NaOH.

Volume of NaOH (ml)	conductance (mS/cm)
0	2.21
1.0	1.84
2.0	1.53
3.0	1.16
4.0	0.83
5.0	0.87
6.0	1.15
7.0	1.46
8.0	1.83
9.0	2.06

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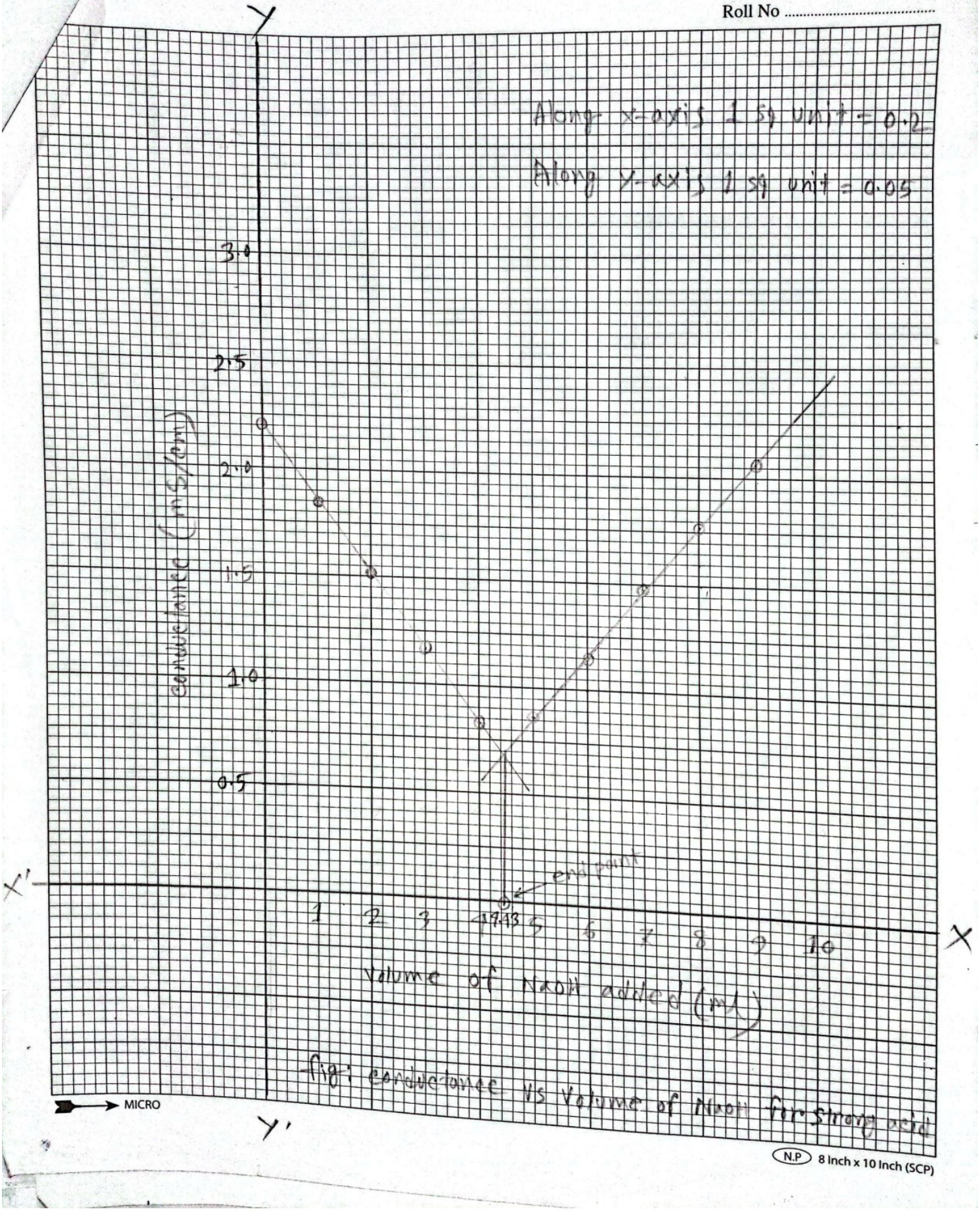

fig: conductance Vs Volume of NaOH for strong acid

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Table-6: conductometric titration of weak acid CH_3COOH with NaOH.

Volume of NaOH (ml)	conductance (mS/cm)
0.0	0.16
0.2	0.19
0.9	0.11
0.6	0.13
0.8	0.15
1.0	0.18
2.0	0.31
3.0	0.99
4.0	0.57
5.0	0.89
6.0	1.10
7.0	1.35
8.0	1.61
9.0	1.87

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NAME :

ROLL :

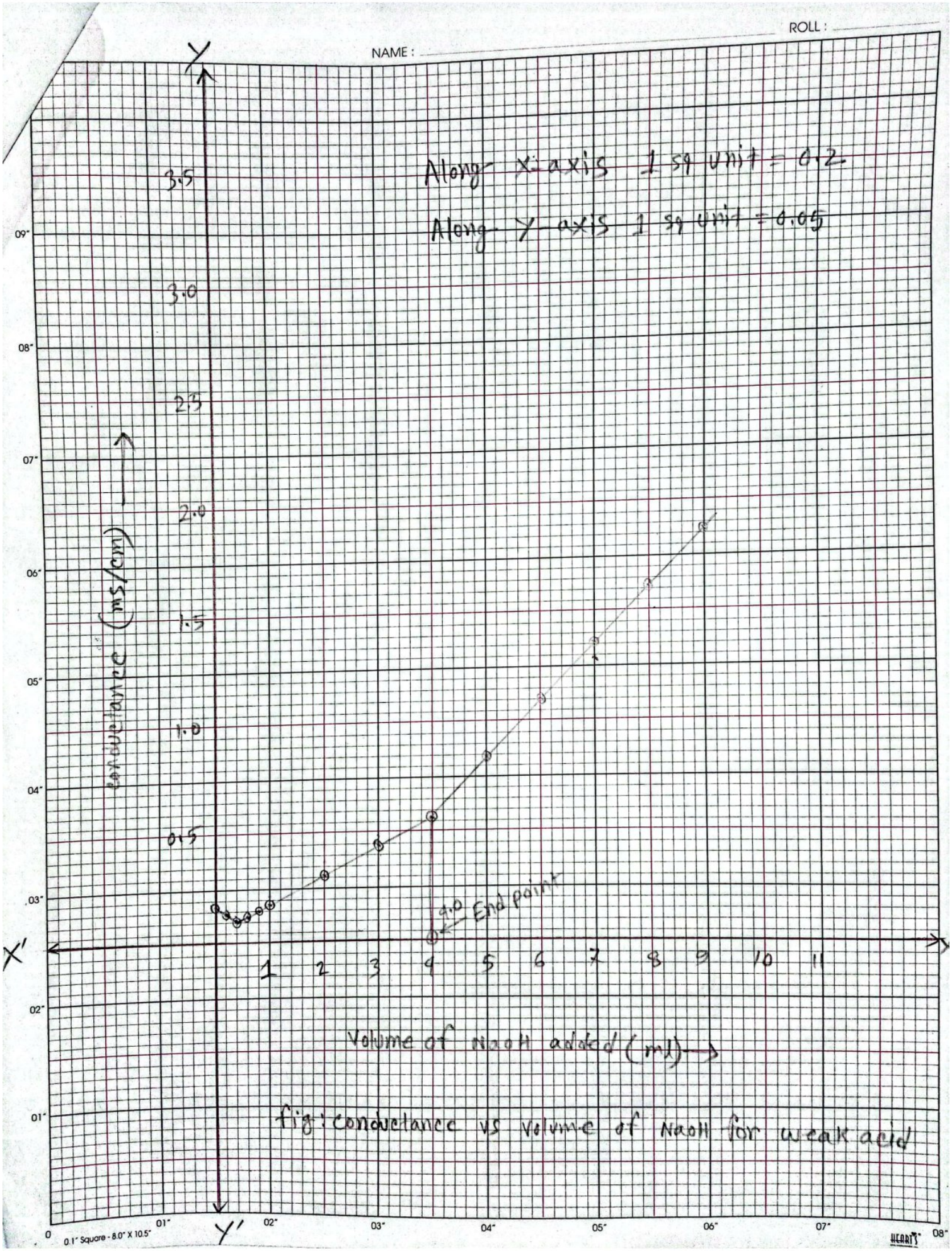

fig: Conductance vs Volume of NaOH for weak acid

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CALCULATION:

Standarisation of 0.2 NaoH against standard 0.1N succinic acid.

No. of Obs.	Volume of 0.1N succinic acid (ml)	Volume of 0.2N NaoH			Mean (ml)
		I.B.R. (ml)	F.B.R. (ml)	Difference (ml)	
1	10.00	0.00	11.10	11.10	
2	10.00	11.10	22.30	11.20	11.20
3	10.00	23.00	34.20	11.20	

Reaction involved:

$$\begin{matrix} CH_2COOH \\ | \\ CH_2COOH \end{matrix} + 2NaOH \longrightarrow \begin{matrix} CH_2-COONa \\ | \\ CH_2-COONa \end{matrix} + 2H_2O$$

NOW,

$$\frac{V_A S_A}{V_B S_B} = \frac{a}{b}$$

$$\therefore S_B = \frac{b V_A S_A}{a V_B}$$

$$= \frac{2 \times 10 \times 0.1}{1 \times 11.20}$$

$$= 0.179 \text{ N}$$

Here,

Volume of acid, $V_A = 10.00$ ml

Strength of acid, $S_A = 0.1$ N

Volume of Base, $V_B = 11.20$ ml

Strength of Base, $S_B = ?$

$a = 1, b = 2$

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For strong acid:

$NaOH + HCl \longrightarrow NaCl + H_2O$

$$\frac{V_A S_A}{V_B S_B} = \frac{a}{b}$$

$$\Rightarrow S_A = \frac{a V_B S_B}{b V_A} = \frac{1 \times 4.43 \times 0.179}{1 \times 10}$$

$$\therefore S_A = 0.079 \text{ N}$$

Here,
Volume of acid, $V_A = 10$ ml
Strength of acid = ?
Volume of NaOH from the graph-1, $V_B = 4.43$ ml
Strength of NaOH, $S_B = 0.179$ N
$a = b = 1$

For weak acid:

$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$

$$S_A = \frac{a V_B S_B}{b V_A} = \frac{1 \times 4.00 \times 0.179}{1 \times 10}$$

$$\therefore S_A = 0.072 \text{ N}$$

Here,
Volume of acid, $V_A = 10$ ml
Strength of acid S_A = ?
Volume of base from the group-2, $V_B = 4.00$ ml
Strength of NaOH
$S_B = 0.179$ N
$a = b = 1$

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RESULT:

1. Strength of Strong acid, HCl is = 0.079 N
2. Strength of weak acid, CH_3COOH is = 0.072 N

DISCUSSION:

1. Apparatus was cleaned by distilled water carefully.
2. Measurement was taken carefully and plotted the graph accurately.
3. NaOH is secondary standard substance so at first it was standarized by 0.1 M succinic acid.
4. when NaOH was added the conductivity of HCl decreased due to neutralization of highly mobile H^+ ions by OH^- ions of NaOH until an equivalent point was reached.
5. on the addition of Base to CH_3COOH, at first conductivity decreased due to common ion acetate and very soon the conductivity was increased due to formation of strong electrolyte CH_3COONa.
6. we titrate HCl and CH_3COOH with NaOH very carefully.

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Jahangirnagar University
Department of Chemistry

Lab Report

Course Title: Physical Chemistry Practical (III)

Course No: CHEM - 315

Experiment No: 11

Experiment Name: Potentiometric titration of (a) strong and (b) weak acids and drawing their neutralisation curves.

Submitted To

Dr. Mohammed Abdullah Khan
Professor
Department of Chemistry
Jahangirnagar University
Savar, Dhaka-1342

Submitted By:

Name: Md Rajib Mia
Roll: 200
Group No: 09
Session: 2020-2021
Department of Chemistry
Jahangirnagar University
Savar, Dhaka-1342

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Experiment NO: 11

Name of experiment: potentiometric titration of (a) strong and (b) week acids by a strong base and to draw their neutralization curves.

Theory: A reaction, in which H^+ ion are produced or consumed, can be followed by observing the pH of the reaction.

$$pH = -\log[H^+]$$

potentiometric titration can serve to determine the equivalence point of titration. The pH will vary slightly with addition of base. However, near the equivalence point, concentration of the H^+ cations fall rapidly.

Apparatus and chemical required:

pH meter, volumetric flask, beakers, strong acid, weak acid, 0.1N KCl solution, standard 0.1N NaOH solution, quinhydrone.

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Date: 8.05.2024
Roll: 200
No. of group: 09

Experiment NO: 11

Experiment Name: potentiometric titration of (a) strong and (b) weak acids by a strong base and drawing their neutralization curves.

DATA:

Table-a: potentiometric titration of strong acid HCl with NaOH.

Volume of NaOH (mL)	pH
0	2.20
1.0	2.22
2.0	2.30
3.0	2.40
4.0	2.46
5.0	2.52
6.0	2.60
7.0	2.75
8.0	8.11
9.0	8.60
10.0	8.79
11.0	8.83
12.0	9.00
13.0	9.11

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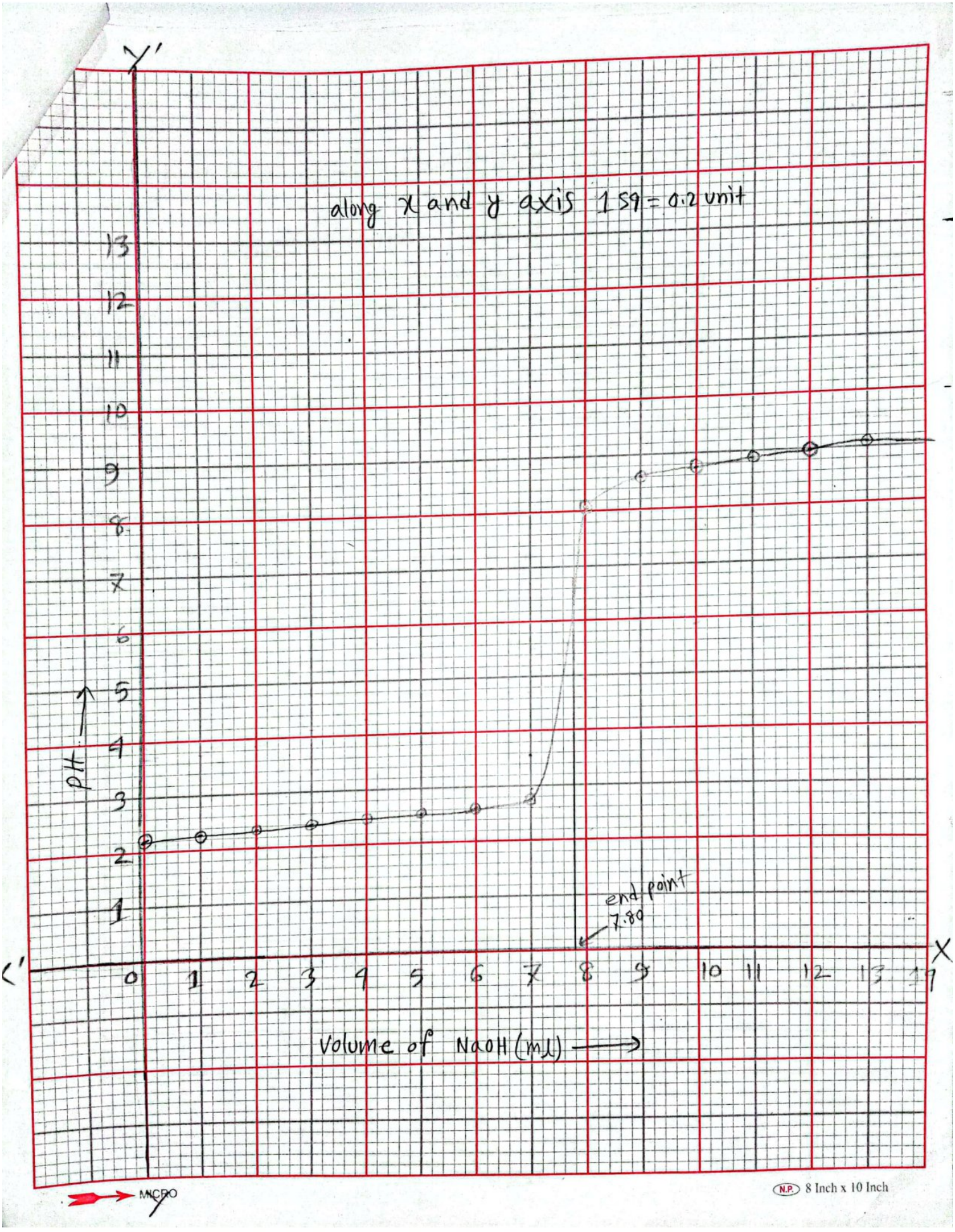

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Table - b: potentiometric titration of weak acid CH_3COOH with NaOH

Volume of NaOH (mL)	pH
0.0	3.69
1.0	3.80
2.0	3.89
3.0	4.10
4.0	4.19
5.0	4.20
6.0	4.30
7.0	4.41
8.0	4.59
9.0	4.91
10.0	5.81
11.0	8.91
12.0	9.49
13.0	9.61
14.0	9.92
15.0	10.01

14/5/25

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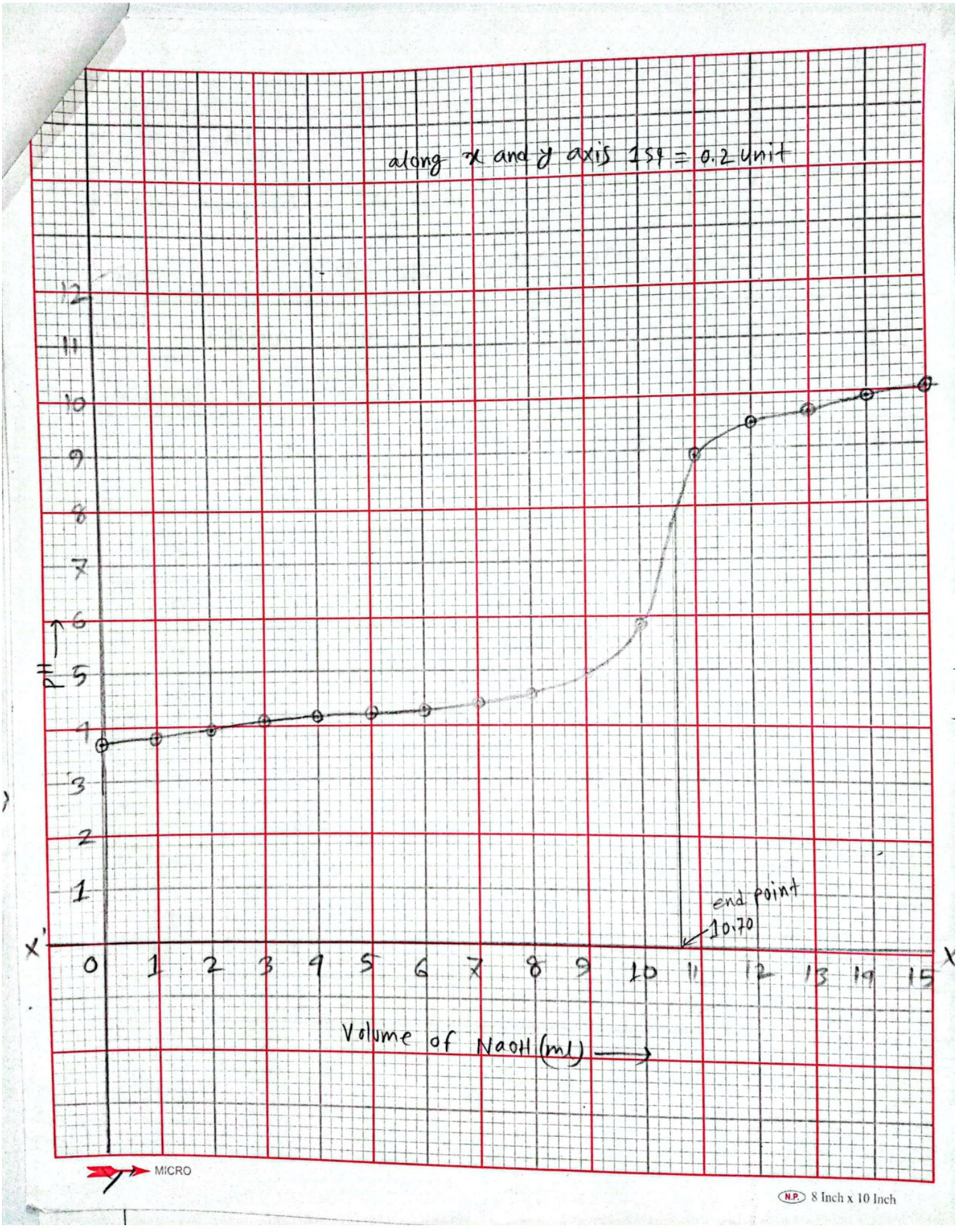

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CALCULATION:

Table-1, Data for drawing graph $\frac{\Delta PH}{\Delta V}$ Vs mean volume of NaOH for strong acid.

Volume of added NaOH(ml)	Mean volume $V_m = \frac{V_1 + V_2}{2}$	$\Delta V = V_2 - V_1$	PH	ΔpH	$\frac{\Delta PH}{\Delta V}$
1.0	0.5	1.0	2.22	0.02	0.02
2.0	1.5	1.0	2.30	0.08	0.08
3.0	2.5	1.0	2.40	0.10	0.10
4.0	3.5	1.0	2.46	0.06	0.06
5.0	4.5	1.0	2.52	0.06	0.06
6.0	5.5	1.0	2.60	0.08	0.08
7.0	6.5	1.0	2.75	0.15	0.15
8.0	7.5	1.0	8.11	5.36	5.36
9.0	8.5	1.0	8.60	0.49	0.49
10.0	9.5	1.0	8.74	0.14	0.14
11.0	10.5	1.0	8.83	0.09	0.09
12.0	11.5	1.0	9.00	0.17	0.17
13.0	12.5	1.0	9.11	0.11	0.11

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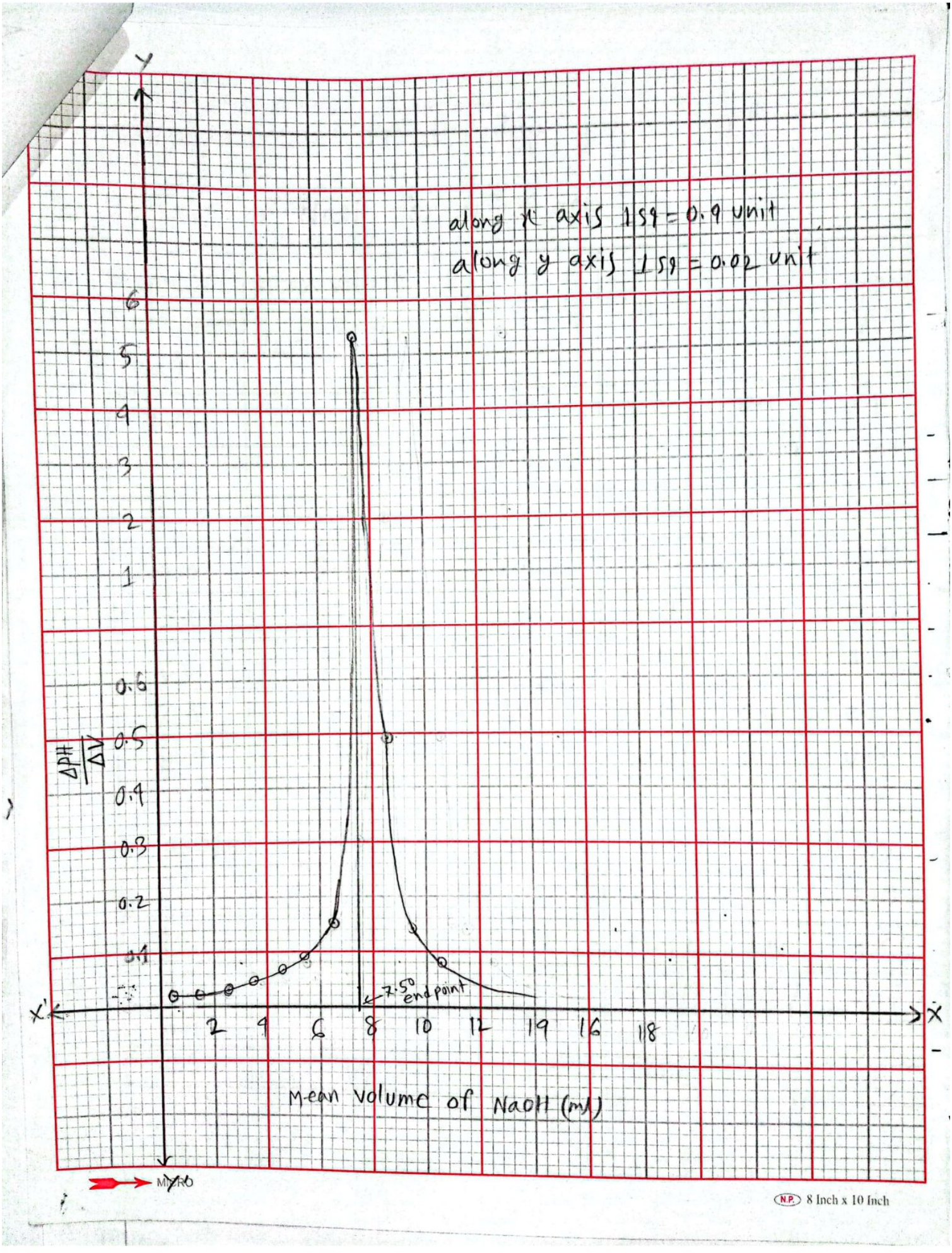

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Table-2: Data for drawing graph $\frac{\Delta pH}{\Delta V}$ vs mean Volume of NaOH for weak acid.

Volume of NaOH added (ml)	Mean volume $V_m = \frac{V_1+V_2}{2}$	$\Delta V = V_2 - V_1$	pH	ΔpH	$\frac{\Delta pH}{\Delta V}$
1.0	0.5	1.0	3.80	0.11	0.11
2.0	1.5	1.0	3.89	0.09	0.09
3.0	2.5	1.0	4.10	0.21	0.4
4.0	3.5	1.0	4.19	0.09	0.09
5.0	4.5	1.0	4.200	0.01	0.01
6.0	5.5	1.0	4.30	0.10	0.10
7.0	6.5	1.0	4.41	0.11	0.11
8.0	7.5	1.0	4.59	0.18	0.18
9.0	8.5	1.0	4.91	0.32	0.32
~~9.9~~ 10.0	9.5	1.0	5.81	0.9	0.9
11.0	10.5	1.0	8.91	2.9	2.9
12	11.5	1.0	9.44	0.53	0.53
13	12.5	1.0	9.61	0.17	0.17
14	13.5	1.0	9.92	0.31	0.31
15	14.5	1.0	10.01	0.09	0.09

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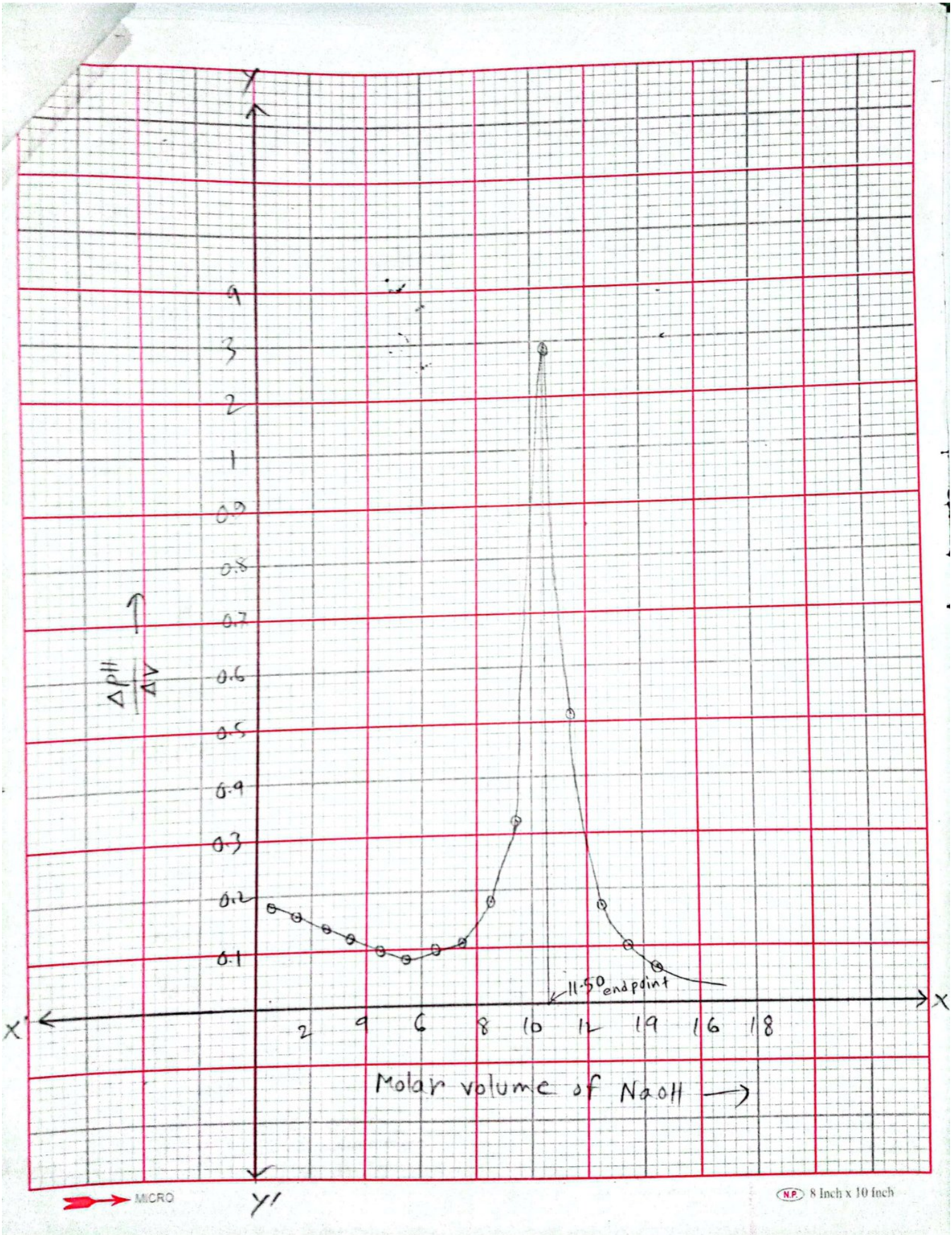

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Standarisation of NaOH solution against standard Succinic acid.

No. of Obs.	Volume of 0.1N succinic acid	Volume of 0.1N NaOH			Mean
		I.B.R (ml)	F.B.R. (ml)	Diff. (ml)	
1	10.00	0.00	19.20	19.20	
2	10.00	20.00	39.10	19.10	19.10
3	10.00	0.00	19.10	19.10	

Reaction involved

$$\begin{matrix} CH_2COOH \\ | \\ CH_2COOH \end{matrix} + 2NaOH \longrightarrow \begin{matrix} CH_2COONa \\ | \\ CH_2COONa \end{matrix} + 2H_2O$$

$$\frac{V_A S_A}{V_B S_B} = \frac{a}{b}$$

$$\therefore S_B = \frac{b\, V_A \cdot S_A}{a\, V_B}$$

$$= \frac{2 \times 10 \times 0.1}{1 \times 19.10}$$

$$= 0.109N$$

$V_A = 10ml$

$S_A = 0.1N$

$V_B = 19.10 ml$

$S_B = ?$

$a = 1, b = 2$

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(a) Titration of Strong acid by NaOH.

$NaOH + HCl \longrightarrow NaCl + H_2O$

$$S_A = \frac{a S_B V_B}{b V_A}$$

$$= \frac{1 \times 0.109 \times 7.50}{1 \times 10}$$

$$= 0.079\ N$$

$V_A = 10\ ml$

$a = 1, b = 1$

$S_A = ?$

$S_B = 0.109\ N$

$V_B = 7.50\ ml$

(b) Titration of weak acid by NaOH

$NaOH + CH_3COOH \longrightarrow CH_3COONa + H_2O$

$$S_A = \frac{a V_B S_B}{b V_A}$$

$$= \frac{1 \times 11.50 \times 0.109}{1 \times 10}$$

$$= 0.1196\ N$$

$V_A = 10\ ml$

$S_A = ?$

$S_B = 0.109\ N$

$V_B = 11.50\ ml$

$a = b = 1$

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RESULT:

(i) Concentration of strong acid, HCl is 0.079N.

(ii) Concentration of weak acid, CH_3COOH is 0.1196N.

(iii) Neutralization curves of strong acid and weak acid are shown in graph-1 and graph-2 respectively.

DICUSSION:

1. NaOH is secondary standard substance, so, it was standarized first.
2. The pH of end point
3. The potentiometric titration is more reliable than volumetric titration.
4. The readings were taken carefully and plotted at graph carefully.
5. The titration were continued till the end point, indicated by a sharp change in the value of pH of the solution.

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Jahangirnagar University
Department of Chemistry

Lab Report

Course Title: Physical Chemistry Practical (III)

Course No: CHEM - 315

Experiment No: 12

Experiment Name: Determination of the the standard electrode potential of zinc and copper.

Submitted To

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Session: 2020-2021
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Experiment NO: 12

Experiment Name: Determination the standard electrode potential of zinc and cupper.

Theory: The saturated calomel electrode (SCE) is connected with copper or zinc electrode and the e.m.f. of the cell is determined by means of potentiometer. Thus

$$E_{cell} = E_{cathode(red)} - E_{anode(red)}$$

Two cells can be represented as:

$Zn, Zn^{2+}(aq)$ || calomel (saturated)

calomel (saturated) || $Cu^{2+}(aq), Cu$

The standard electrode potential (E°) is given by;

$$E = E^\circ - \frac{2.303RT}{nF} \log Q$$

Apparatus and chemical required:

potentiometer, standard cell, galvanometer, standard calomel electrode, zinc electrode, copper electrode 0.1M Solution $ZnSO_4$ and $CuSO_4$.

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Date: 14.05.2024
Roll: 200
Group: 09

Experiment NO: 12

Name of the experiment: Determination of the standard electrode potential of zinc and copper.

DATA:

Table-a: E.m.f of $ZnSO_4$ solution with saturated calomel electrode.

concentration of $ZnSO_4$ (M)	emf, E_{cell} (V)
0.0001 M	0.862
0.001M	0.920
0.01M	0.976
0.1M	0.989

Table-b: Emf of $CuSO_4$ Solution with saturated calomel electrode.

concentration of $CuSO_4$ (M)	e.mf, E_{cell} (V)
0.0001M	0.049
0.001M	0.060
0.01M	0.073
0.1M	0.083

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CALCULATION:

Table-1: Data for Zn - SCE cell

concentration of $ZnSO_4$ solution(M)	E.M.F of cell E_{cell}(V)	Mean activity of efficient γ_+ at 25°C	$M_{Zn^{2+}}.\gamma_+$	$\log(M_{Zn^{2+}}.\gamma_+)$
0.1	0.989	0.161	0.0161	-1.79
0.01	0.976	0.900	0.009	-2.40
0.001	0.920	0.431	0.000431	-3.37
0.0001	0.862	0.432	0.0000432	-4.36

Table-2: Data for copper-SCE cell

concentration of $CuSO_4$ solution(M)	E.M.F of cell E_{cell}(V)	Mean activity of efficient γ_+ at 25°C	$M_{Cu^{2+}}.\gamma_+$	$\log(M_{Cu^{2+}}.\gamma_+)$
0.1	0.083	0.158	0.0158	-1.80
0.01	0.073	0.409	0.00409	-2.39
0.001	0.060	0.432	0.000432	-3.36
0.0001	0.049	0.437	0.0000437	-4.36

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calculation of $E^{\circ}_{Zn/Zn^{2+}}$:

$$E_{cell} = E_{Zn/Zn^{2+}} + E_{calomel(red)}$$

$$\Rightarrow E_{cell} = E^{\circ}_{Zn/Zn^{2+}} - \frac{2.303RT}{2F} \log(M_{Zn^{2+}} \cdot \gamma_{\pm}) + E_{cal(red)}$$

$$\therefore E^{\circ}_{Zn/Zn^{2+}} = E_{cell} + \frac{2.303RT}{2F} \log(M_{Zn^{2+}} \cdot \gamma_{\pm}) - E_{cal(red)}$$

Redox potential of saturated calomel electrode = 0.244V at 25°C.

1. For 0.1M $ZnSO_4$ solution:

$$E^{\circ}_{Zn/Zn^{2+}} = 0.989 + \frac{2.303 \times 8.314 \times 298}{2 \times 96500} \times (-1.79) - 0.244$$

$$= 0.69\,V$$

2. For 0.01M $ZnSO_4$ solution:

$$E^{\circ}_{Zn/Zn^{2+}} = 0.976 + \frac{2.303 \times 8.314 \times 298}{2 \times 96500} \times (-2.40) - 0.244$$

$$= 0.66\,V$$

3. For 0.001 M $ZnSO_4$ solution:

$$E^{\circ}_{Zn/Zn^{2+}} = 0.920 + \frac{2.303 \times 8.314 \times 298}{2 \times 96500} \times (-3.37) - 0.244$$

$$= 0.58\,V$$

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4. For 0.0001 M $ZnSO_4$ solution :

$$E^{\circ}_{Zn/Zn^{2+}} = 0.862 + \frac{2.303 \times 8.314 \times 298}{2 \times 96500} \times (-4.36) - 0.299$$

$$= 0.57 \text{ V}$$

calculation for $E_{Cu^{2+}/Cu}$:

$$E_{cell} = E_{cal(ox)} - E_{Cu/Cu^{2+}}$$

$$= E_{cal(ox)} - E^{\circ}_{Cu/Cu^{2+}} + \frac{2.303RT}{2F} \log(M_{Cu^{2+}} \cdot \gamma^{+})$$

$$\therefore E^{\circ}_{Cu^{2+}/Cu} = E_{cell} - \frac{2.303RT}{2F} \log(M_{Cu^{2+}} \cdot \gamma^{+}) + E_{cal(Red)}$$

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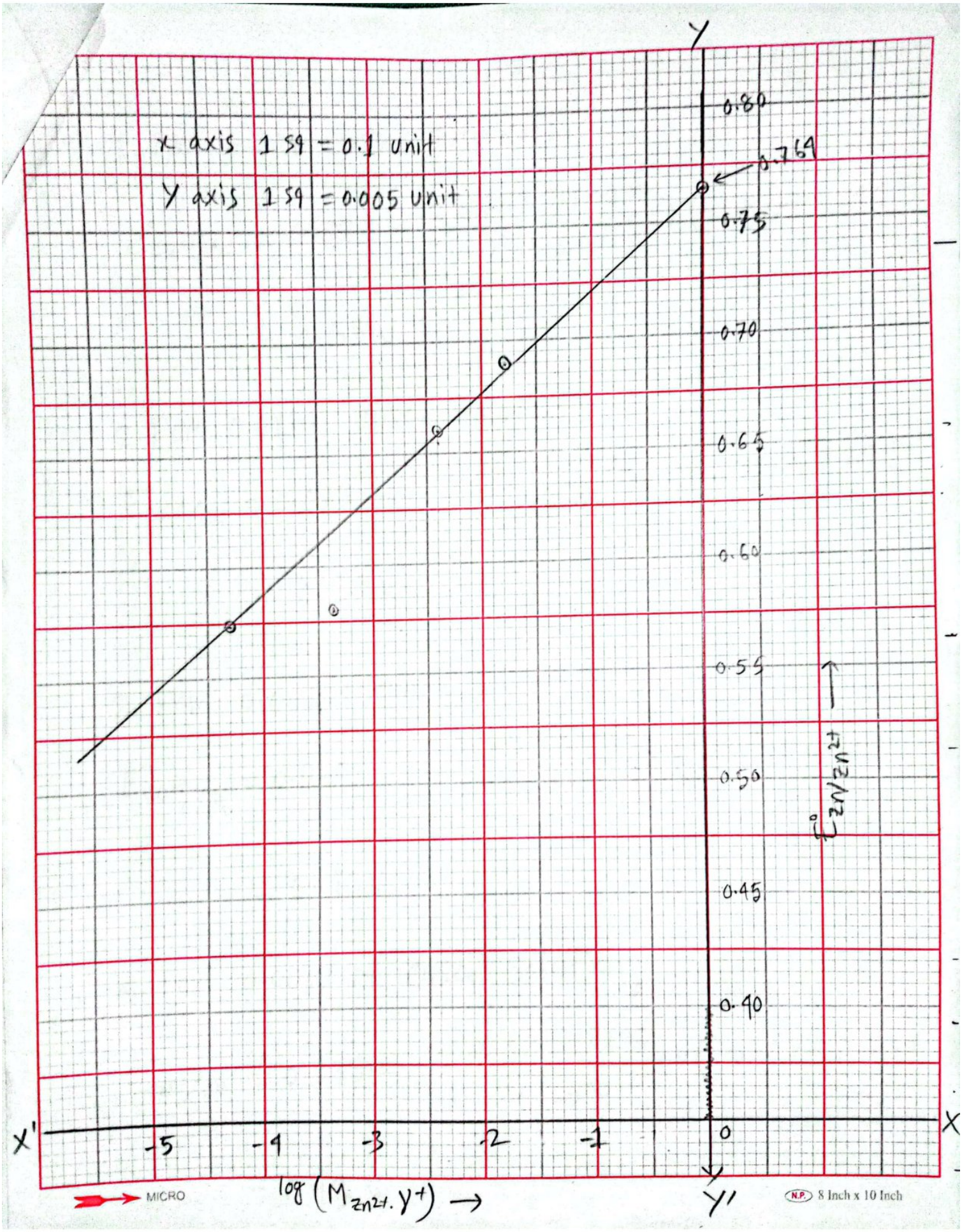

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1. For 0.1M $CuSO_4$ solution:

$$E^{\circ}_{Cu^{2+}/Cu} = 0.083 - \frac{2.303 \times 8.314 \times 298}{2 \times 96500} \times (-1.80) + 0.244$$

$$= 0.38\ V$$

2. For 0.01M $CuSO_4$ solution:

$$E^{\circ}_{Cu^{2+}/Cu} = 0.073 - \frac{2.303 \times 8.314 \times 298}{2 \times 96500} \times (-2.39) + 0.244$$

$$= 0.39\ V$$

3. For 0.001M $CuSO_4$ solution:

$$E^{\circ}_{Cu^{2+}/Cu} = 0.060 - \frac{2.303 \times 8.314 \times 298}{2 \times 96500} \times (-3.36) + 0.244$$

$$= 0.40\ V$$

4. For 0.0001 M $CuSO_4$ solution:

$$E^{\circ}_{Cu^{2+}/Cu} = 0.049 - \frac{2.303 \times 8.314 \times 298}{2 \times 96500} \times (-4.36) + 0.244$$

$$= 0.43\ V$$

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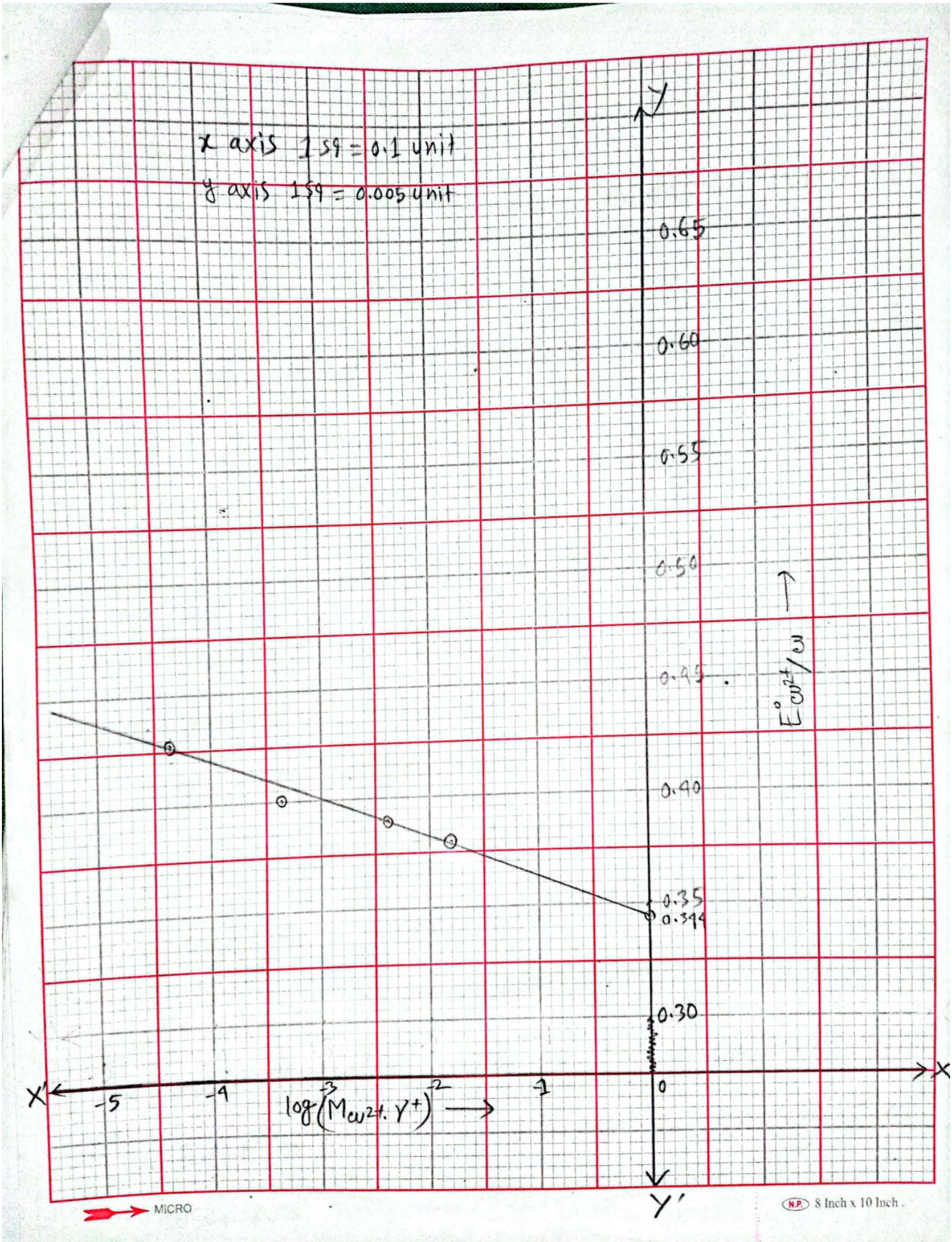

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RESULT:

1. From graph-1, standard oxidation potential of Zn is 0.769 V.
2. From graph-2, standard reduction potential of Cu is 0.344 V.

DISCUSSION!

1. Solutions of different concentration was prepared carefully.
2. The cell was connected accurately.
3. Reading taken carefully
4. The graph was plotted preciously as possible.
5. The salt bridge washed to place different concentration solutions.
6. potentiometer readings has taken carefully.
7. E.M.F. of cell using different concentrations of $CuSO_4$ and $ZnSO_4$ measured carefully.

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